Ordinal Googology

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01 Introduction

When I was learning Googology, I was never really interested in naming specific large numbers, but instead learning about *how* people generated large numbers. I tried understanding notation like B.E.A.F. and BAN but the rules were complex and they felt too clunky and inelegant to deal with. I then (re-)discovered about the Fast-Growing Hierarchy and how it can systematically categorize the growth of sequences, and it let me on a rabbit hole to finally be able to grasp the growth rate of $\mathrm{TREE}(n)$. However, when I tried digging into the formal definitions of the things I was writing, I realized that there was a lot that I wasn't sure of myself. If I want to compute something like $f_{\Gamma_0}(3)$, what exactly am I supposed to do? There's a Geometry Dash level where I have to click $f_{\psi_0(\Omega_\omega)}(520)$ times, what does that mean? This book is a compilation of my attempts to understand the math involved in googology.

Generally, I introduce new concepts more informally, then go back around to formalize them, though I feel in some sections I might have frontloaded the formality too much.

Feel free to correct any mistakes or re-work some sections. You may also suggest or contribute chapters for new topics, though I'd like it to fit within the general "skill level" of the book. Some concepts that I am currenty nowhere confident enough to write about are:

- Inaccessible cardinals, Mahlo Cardinals, Weakly Compact Cardinals, Stability, Reflection and whatever OCF can use them
- y sequence
- Laver tables
- Proof Theoretic Ordinals (though ordinal analysis might be way too technical to be summarized in a chapter)

References:
- Googology wiki(s)
- The googology discord
- An Introduction to Proof Theory: Normalization, Cut-elimination, and Consistency Proofs by Paolo Mancosu, Richard Zach, and Sergio Galvan
- Zhongshu Wu PrSS document on google drive

1 Prerequisites

- Set theory
  - Notations $\cup, \cap, \in, \subset, \setminus, \emptyset, \times$ (cartesian product)
  - Notation $\{x \mid P(x)\}$ where $P(x)$ is a statement about x
  - Set vs class
- The Natural Numbers $\mathbb{N}$ (in this book we include $0 \in \mathbb{N}$)
- Knowledge on functions

02 Hyperoperators

1 Addition, Multiplication, and Exponentiation

Before we get to large numbers with powerful operations, let's first start from concepts we are more familiar with. Let's first start with the very basic **Successor Function**.

Definition 1

The **Successor Function** is the function $S(x) =$ the least natural number greater than x, or more commonly, $S(x) = x + 1$

In this case, **Addition** is actually just the Successor Function applied repeatedly:

Definition 2

Addition can be defined recursively as:
1. $a + 0 = a$
2. $a + S(b) = S(a + b)$

Alternatively, it can also be seen as the Successor Function applied repeatedly:

$$a + b = \underbrace{S(S(...S(a)))}_{S() \text{ applied } b \text{ dot}} = a + \underbrace{1 + 1 + ... + 1}_{b \text{ number of } +1}$$

Example 1

Evaluate $2 + 3$ with the recursive definition:

$$\begin{aligned} 2 + 0 &= 2 \\ 2 + 1 &= 2 + S(0) = S(2 + 0 = 2) = 3 \\ 2 + 2 &= 2 + S(1) = S(2 + 1 = 3) = 4 \\ 2 + 3 &= 2 + S(2) = S(2 + 2 = 4) = 5 \end{aligned}$$

Note that each line requires the previous $2 + n$ to be known before the Successor Function $S(x)$ can be applied onto it

We next move onto **Multiplication**, which, as you have learnt in school, is repeated addition:

Definition 3

Multiplication can be defined recursively as:
1. $a \cdot 0 = 0$
2. $a \cdot (b + 1) = (a \cdot b) + a$

or alternatively, as just repeated addition:

$$a \cdot b = \underbrace{a + a + ... + a}_{a \text{ is here } b \text{ times}}$$

Example 2

Evaluate $2 \cdot 3$

- With the recursive definition:

$$\begin{aligned} 2 \cdot 0 &= 0 \\ 2 \cdot 1 &= (2 \cdot 0) + 2 = 2 \\ 2 \cdot 2 &= (2 \cdot 1) + 2 = 4 \\ 2 \cdot 3 &= (2 \cdot 2) + 2 = 6 \end{aligned}$$

- or just applying addition repeatedly:

$$2 \cdot 3 = \underbrace{2 + 2 + 2}_{2 \text{ is here } 3 \text{ times}} = 6$$

Now let's move on to **Exponentiation**, which is repeated multiplication:

Definition 4

Exponentiation can be defined recursively as such:

$$a^b := \begin{cases} 1 & (b = 0) \\ a^{b-1} \cdot a & (b > 0) \end{cases}$$

or alternatively, as just repeated multiplication: $a^b = \underbrace{a \cdot a \cdot a \cdot \ldots \cdot a}_{a \text{ is here } b \text{ times}}$

Exponentiation may also be denoted using Donald Knuth's **up-arrow notation** as $a \uparrow b = a^b$

Example 3

Evaluate 2^3

- With the recursive definition:

$$\begin{aligned} 2^0 &= 1 \\ 2^1 &= (2^0) \cdot 2 = 2 \\ 2^2 &= (2^1) \cdot 2 = 4 \\ 2^3 &= (2^2) \cdot 2 = 8 \end{aligned}$$

- or just applying multiplication repeatedly:

$$2 \cdot 3 = \underbrace{2 \cdot 2 \cdot 2}_{2 \text{ is here } 3 \text{ times}} = 8$$

We can start to see a pattern here, where each new, more powerful operator is defined as repeatedly applying the previous operator. So what happens when we repeatedly do this for exponentiation?

2 Tetration and Beyond

Definition 5

Tetration is defined as repeated exponentiation. This can be expressed recursively:

$$a \uparrow\uparrow 1 = a$$
$$a \uparrow\uparrow b = a^{a \uparrow\uparrow (b-1)} = a \uparrow (a \uparrow\uparrow (b-1))$$

or simply as repeated exponentiation, but computed from right to left:

$$a \uparrow\uparrow b = \underbrace{a^{a^{a^{\cdot^{\cdot^{\cdot}}}}}}_{a \text{ is here } b \text{ times}} = \underbrace{a \uparrow a \uparrow \ldots \uparrow a}_{a \text{ is here } b \text{ times}}$$

Example 4

Evaluate $2 \uparrow\uparrow 3$

- With the recursive definition:

$$2 \uparrow\uparrow 1 = 2$$
$$2 \uparrow\uparrow 2 = 2 \uparrow (2 \uparrow\uparrow 1) = 2^2 = 4$$
$$2 \uparrow\uparrow 3 = 2 \uparrow (2 \uparrow\uparrow 2) = 2^4 = 16$$

- or just applying exponentiation repeatedly:

$$2 \uparrow\uparrow 3 = \underbrace{2 \uparrow 2 \uparrow 2}_{2 \text{ is here } 3 \text{ times}} = 2^{2^2} = 2^{(2^2)} = 2^4 = 16$$

We can similarly define **Pentation**:

Definition 6

Pentation is repeated Tetrations like so:

$$a \uparrow\uparrow\uparrow b = \underbrace{a \uparrow\uparrow a \uparrow\uparrow \ldots \uparrow\uparrow a}_{a \text{ is here } b \text{ times}}$$

Example 5

$$\begin{aligned}
2 \uparrow\uparrow\uparrow 3 = \underbrace{2 \uparrow\uparrow 2 \uparrow\uparrow 2}_{2 \text{ is here } 3 \text{ times}} &= 2 \uparrow\uparrow (2 \uparrow\uparrow 2) \\
&= 2 \uparrow\uparrow (2^2) \\
&= 2 \uparrow\uparrow 4 \\
&= 2 \uparrow 2 \uparrow 2 \uparrow 2 \\
&= 2 \uparrow 2 \uparrow (2^2) \\
&= 2 \uparrow 2 \uparrow 4 \\
&= 2 \uparrow (2^4) \\
&= 2 \uparrow 16 \\
&= 65536
\end{aligned}$$

Now, 65536 might not look like much, but when you put in slightly larger numbers like $3 \uparrow\uparrow\uparrow 3$, you will get monstrous results:

Example 6

$$\begin{aligned} 3 \uparrow\uparrow\uparrow 3 &= 3 \uparrow\uparrow 3 \uparrow\uparrow 3 \\ &= 3 \uparrow\uparrow \left(3^{3^3}\right) \\ &= 3 \uparrow\uparrow (\sim 7.63 \times 10^{12}) \\ &= \underbrace{3 \uparrow 3 \uparrow \ldots \uparrow 3}_{\sim 7.63\times 10^{12} \text{ arrows}} \end{aligned}$$

We can generalize the definition of increasingly powerful hyperoperators as such:

Definition 7

Let $a \uparrow^n b = a\underbrace{\uparrow\uparrow \ldots \uparrow}_{n}b$

$a \uparrow^{n+1} b = \underbrace{a \uparrow^n a \uparrow^n \ldots \uparrow^n a}_{a \text{ is here } b \text{ times}}$

With up-arrow notation, we can already make enormously big numbers like Graham's number:

Example 7

$g_0 = 4$

$g_{n+1} = 3 \uparrow^{g_n} 3 = 3\underbrace{\uparrow\uparrow\uparrow \ldots \uparrow}_{g_n \text{ times}}3$

So for example:

$g_1 = 3 \underbrace{\uparrow\uparrow\uparrow\uparrow}_{4} 3$

$g_2 = 3 \underbrace{\uparrow\uparrow\uparrow \ldots \uparrow}_{3\uparrow\uparrow\uparrow\uparrow 3 \text{ times}} 3$

Graham's number is defined as g_{64}

03 Introduction to Ordinals

1 Informal definition

The easiest way to think about ordinals is as an extension of the natural numbers $\mathbb{N} = \{0, 1, 2, 3, ...\}$ (We include 0 in the natural numbers for this book).

Definition 8

We define ω as the **first ordinal larger than all the natural numbers**, or **the first transfinite ordinal**.

We can continue by having an $\omega + 1, \omega + 2, ...$, so we have something like $\{0, 1, 2, ..., \omega, \omega + 1, \omega + 2, ...\}$

We then have $\omega + \omega = \omega \cdot 2, \omega \cdot 3, \omega \cdot 4, ...$ until we get $\omega \cdot \omega = \omega^2, \omega^3, \omega^\omega, \omega^{\omega+1}, \omega^{\omega^\omega}, ...$ Eventually, we reach $\omega^{\omega^{\cdot^{\cdot^{\cdot}}}} = \varepsilon_0$. The special property of ε_0 is that $\omega^{\varepsilon_0} = \varepsilon_0$, as

$$\omega^{\varepsilon_0} = \omega^{\left(\omega^{\omega^{\cdot^{\cdot^{\cdot}}}}\right)} = \omega^{\omega^{\omega^{\cdot^{\cdot^{\cdot}}}}} = \varepsilon_0$$

We note that there are many intermediate ordinals that, while not mentioned explicitly, are still present, like $\omega^{\omega^3} < \omega^{\omega^3+\omega^2\cdot 3+\omega\cdot 4+5} + 6 < \omega^{\omega^4}$.

2 Von Neumann definition of ordinals

Definition 9

An ordinal is defined as **the set of all ordinals lesser than that ordinal**.

Let's look at some examples of how this plays out:

- There are no ordinals smaller than 0, so $0 = \emptyset = \{\}$, the empty set.
- For the ordinal 1, 0 is the only ordinal smaller than 1, so $1 = \{0\} = \{\emptyset\} = \{\{\}\}$.
- Similarly, we have $2 = \{0, 1\} = \{\emptyset, \{\emptyset\}\}$, $3 = \{0, 1, 2\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$, etc...
- In this definition, ω is defined as the set of all ordinals less than ω, i.e. all natural numbers:

$$\omega = \{0, 1, 2, ...\} = \mathbb{N}$$

- $\omega + 1$ can similarly be defined as all the natural numbers + ω:

$$\omega + 1 = \{0, 1, 2, ..., \omega\} = \mathbb{N} \cup \{\omega\} = \omega \cup \{\omega\}$$

- $\omega \cdot 2 = \omega + \omega$ can be defined as the ordinal larger than all $\omega + n$:

$$\omega \cdot 2 = \{0, 1, 2, ..., \omega, \omega + 1, \omega + 2, ...\}$$

We can therefore see from this that **for any set of ordinals**, there exists another ordinal greater than all of them. Now, this can lead to a contradiction - what about the set of all ordinals? (this is Burali-Forti paradox). In modern set theory, however, it is not possible to construct "the set of all

ordinals" which resolves this paradox. We only have the ***proper class* of all ordinals**, denoted **Ord**, **On**, or less commonly, **ON**.

We can see from this definition of ordinals that for ordinals α and β, $\alpha < \beta$ can be defined as $\alpha \in \beta$. Since α also contains all ordinals smaller than itself, $\alpha < \beta$ can be written as $\alpha \subsetneq \beta$ ($\alpha \subset \beta$ and $\alpha \neq \beta$)

3 Ordinal Arithmetic

We previously defined the successor function $S(n)$ to be the least natural number greater than n, or just $n + 1$. Using the von Neumann definition of ordinals, we can also write the successor function as:

Definition 10

The **successor function** $S(\alpha) = \alpha \cup \{\alpha\} = \alpha + 1$

Definition 11

A **Successor Ordinal** β is an ordinal which can be expressed as the **successor of another ordinal**.

A **Limit Ordinal** is an ordinal that is **not a successor ordinal**.

Example 8

Categorize the following ordinals into whether they are limit or successor ordinals:
1. ω
2. $\omega + 5$
3. ω^2

Explanations:
1. ω is bigger than all natural numbers, so there is no natural number whose successor is ω. Therefore ω is a **limit ordinal**.
2. $\omega + 5$ is the successor of $\omega + 4$. Therefore $\omega + 5$ is a **successor ordinal**.
3. ω^2 is the ordinal bigger than all $\omega \cdot n$. Since there is no greatest $\omega \cdot n$, ω^2 can't be a successor to any of them. Therefore ω^2 is a **limit ordinal**.

We previously defined addition recursively here. Let's add a new case to accommodate limit ordinals:

Definition 12

Addition can be defined recursively as:
1. $\alpha + 0 = a$
2. $\alpha + S(\beta) = S(\alpha + \beta)$
3. $\alpha + \beta = \bigcup_{\gamma<\beta} (\alpha + \gamma)$ when β is a limit ordinal

Example 9

Evaluate $\omega + 1$ and $1 + \omega$. Are they the same?

$$\begin{aligned}\omega + 1 &= \omega + S(0) \\ &= S(\omega + 0) \text{ (Rule 2)} \\ &= S(\omega) \text{ (Rule 1)} \\ &= \omega + 1\end{aligned}$$

$$\begin{aligned}1 + \omega &= \bigcup_{\delta<\omega}(1+\delta) \text{ (Rule 3)} \\ &= 1 \cup 2 \cup 3 \cup \ldots \\ &= \{0\} \cup \{0,1\} \cup \{0,1,2\} \cup \ldots \\ &= \{0,1,2,\ldots\} \\ &= \omega\end{aligned}$$

We see that addition is not commutative (i.e. $a + b$ is not always equal to $b + a$)

We can similarly accommodate limit ordinals into our definition of multiplication and exponentiation too:

Definition 13

Multiplication can be defined recursively as:
1. $\alpha \cdot 0 = 0$
2. $\alpha \cdot S(\beta) = (\alpha \cdot \beta) + \beta$
3. $\alpha \cdot \beta = \bigcup_{\gamma<\beta}(\alpha \cdot \gamma)$ when β is a limit ordinal

Example 10

Evaluate $\omega \cdot 2$ and $2 \cdot \omega$

$$\begin{aligned}\omega \cdot 2 &= \omega \cdot S(1) \\ &= (\omega \cdot 1) + \omega \\ &= ((\omega \cdot 0) + \omega) + \omega \\ &= \omega + \omega\end{aligned}$$

$$\begin{aligned}2 \cdot \omega &= \bigcup_{\gamma<\omega}(2 \cdot \gamma) \\ &= 0 \cup 2 \cup 4 \cup \ldots \\ &= \{\} \cup \{0,1\} \cup \{0,1,2,3\} \cup \ldots \\ &= \{0,1,2,3,\ldots\} \\ &= \omega\end{aligned}$$

We see that multiplication is not commutative either (i.e. $a \cdot b$ is not always equal to $b \cdot a$)

Multiplication can also be denoted with the $\times$ symbol (e.g. $\omega \times 2$), or even without any symbol, like in algebra (e.g. $\omega 2$)

Definition 14

Exponentiation can be defined recursively as:
1. $\alpha^0 = 1$

2. $\alpha^{S(\beta)} = \alpha^\beta \cdot \alpha$
3. $\alpha^\beta = \bigcup_{\gamma<\beta} (\alpha^\gamma)$ when β is a limit ordinal

The usual exponential rules apply, e.g. $\alpha^{\beta+\gamma} = \alpha^\beta \cdot \alpha^\gamma$, and $\alpha^{\beta\cdot\gamma} = \left(\alpha^\beta\right)^\gamma$.

Distributivity holds on the left, but not on the right:

$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$$
$$(\alpha + \beta) \cdot \gamma \neq \alpha \cdot \gamma + \beta \cdot \gamma$$

As such we can also "cancel" terms on the left, but no on the right:

$$\alpha \cdot \beta = \alpha \cdot \gamma \Rightarrow \beta = \gamma$$
$$\beta \cdot \alpha = \gamma \cdot \alpha \not\Rightarrow \beta = \gamma$$

If you recall, $1 + \omega = \omega$. We can show a more general version of this:

Proposition 1

If $\alpha < \beta$, $\omega^\alpha + \omega^\beta = \omega^\beta$

Proof

Let $\beta = \alpha + \gamma$ for some ordinal γ. Then:

$$\begin{aligned}
\omega^\alpha + \omega^\beta &= \omega^\alpha + \omega^{\alpha+\gamma} \\
&= \omega^\alpha + \omega^\alpha \cdot \omega^\gamma \\
&= \omega^{\alpha(1+\omega^\gamma)} \\
&= \omega^\alpha \cdot \omega^\gamma \\
&= \omega^{\alpha+\gamma} \\
&= \omega^\beta
\end{aligned}$$

Higher hyperoperators are tricky to define for transfinite ordinals, so we will only allow addition, multiplication and exponentiation on transfinite ordinals.

4 Ordinals versus Cardinals

The natural numbers $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ can be used in 2 ways: denoting the **position** of an element in a sequence, or denoting the **size** of a set.

Denoting the position leads to **Ordinal numbers**, while denoting the size leads to **Cardinal numbers**. For finite sets it both are the same, but when it comes to infinite sets and sequences, there are differences.

Definition 15

Two sets are said to have the same **cardinality** if there is a **one-to-one correspondence (bijection)** between them.

Note that since we have defined ordinals as sets, we can measured the cardinality of ordinals.

Example 11

Show that ω and $\omega 2$ have the same cardinality.

We can construct a bijection f from ω to $\omega 2$ as such:

$$\begin{aligned}
&0 \to 0 \\
&1 \to \omega \\
&2 \to 1 \\
&3 \to \omega + 1 \\
&\text{and in general:} \\
&2n \to n \\
&2n + 1 \to \omega + n
\end{aligned}$$

To be precise, to show that f is a bijection, we need to show that:

1. f is total: $f(n)$ is defined for all $n \in \omega$
2. f is injective: if $n_1 \neq n_2$ then $f(n_1) \neq f(n_2)$
3. f is surjective: for every $\alpha \in \omega 2$ there exists an $n \in \omega$ such that $f(n) = \alpha$

Every natural number is a cardinal. For infinite cardinals, they are labelled as $\aleph_0, \aleph_1, \aleph_2, \ldots$. We set $\aleph_0$ to be the cardinality of ω (and from the above example, this is the cardinality of $\omega 2$ too).

But how do we know that there are cardinalities above $\aleph_0$? We know there are cardinalities larger than $\aleph_0$, as the cardinality of the set of all real numbers $\mathbb{R}$ is $2^{\aleph_0}$, which is greater than $\aleph_0$. But what is $2^{\aleph_0}$ in terms of $\aleph_n$? Is $2^{\aleph_0} = \aleph_1$ (This is known as the Continuum Hypothesis)? It turns out that we can set $2^{\aleph_0} = \aleph_n$ for *any* finite n, and it will be consistent with ZFC.

In ZFC, we assign each cardinal to an ordinal via **von Neumann cardinal assignment**:

Definition 16

Von Neumann cardinal assignment defines the **cardinality of a set** as the **smallest ordinal that has a bijection with the set**.

Therefore by this definition, **cardinals are ordinals** and we can assign cardinals to ordinals like this: $\aleph_0 = \omega, \aleph_1 = \omega_1, \aleph_2 = \omega_2, \ldots$.

Definition 17

We say a set, and by extension ordinal, is **countable**, if its cardinality is less than ω_1 (or $\aleph_1$).

04 The Fast-Growing Hierarchy

Definition 18

The **fast-growing hierarchy** (FGH) is defined as follows:

1. $f_0(n) := S(n) = n + 1$
2. $f_{\alpha+1} := f_\alpha^n(n) := \underbrace{f_\alpha(f_\alpha(...f_\alpha(n)))}_{f_\alpha \text{ applied } n \text{ times}}$
3. $f_\alpha(n) := f_{\alpha[n]}(n)$ if α is a limit ordinal.

$\alpha[n]$ in this case represents the nth term in the fixed fundamental sequence assigned to the ordinal α.

WARNING

The fast-growing hierarchy is **ill-defined** unless a **specific choice of a system of fundamental sequences is explicitly fixed** in the context.

Wait, but what is a fundamental sequence? For now, we will ignore that warning as we will won't be dealing with limit ordinals.

1 The Fast-Growing Hierarchy for finite ordinals

Before we start plugging in infinite ordinals into the FGH, let's try figuring out the growth rate of the first few f_1, f_2, f_3, f_4:

$$f_1(n) = f_0^n(n) = \underbrace{f_0(f_0(...f_0(n)))}_{\text{applied } n \text{ times}} = n + \underbrace{1 + 1 + ... + 1}_{n \text{ times}} = 2n$$

$$f_2(n) = f_1^n(n) = \underbrace{f_1(f_1(...f_1(n)))}_{\text{applied } n \text{ times}} = \underbrace{2 \cdot 2 \cdot ...2}_{n \text{ times}} \cdot n = 2^n \cdot n > 2^n$$

$$f_3(n) = f_2^n(n) = \underbrace{f_2(f_2(...f_2(n)))}_{\text{applied } n \text{ times}} > \underbrace{2 \uparrow 2 \uparrow ... \uparrow 2}_{n \text{ times}} \uparrow n$$

$$> \underbrace{2 \uparrow 2 \uparrow ... \uparrow 2}_{n \text{ times}} = 2 \uparrow\uparrow n$$

$$f_4(n) = f_3^n(n) = \underbrace{f_3(f_3(...f_3(n)))}_{\text{applied } n \text{ times}} > \underbrace{2 \uparrow\uparrow 2 \uparrow\uparrow ... \uparrow\uparrow 2}_{n \text{ times}} \uparrow\uparrow n$$

$$> \underbrace{2 \uparrow\uparrow 2 \uparrow\uparrow ... \uparrow\uparrow 2}_{n \text{ times}} = 2 \uparrow\uparrow\uparrow n$$

We see that for every new f_m, we get increasingly powerful hyperoperators:

FGH function	corresponding growth rate
$f_1(n)$	$2n$ (Multiplication-level)
$f_2(n)$	$> 2^n$ (Exponentiation-level)
$f_3(n)$	$> 2 \uparrow\uparrow n$ (Tetration-level)

FGH function	corresponding growth rate
$f_4(n)$	$> 2 \uparrow\uparrow\uparrow n$ (Pentation-level)
$f_m(n)$	$> 2 \uparrow^{m-1} n$ ($(m-1)$th hyperoperator-level)

2 Fast-Growing Hierarchy for ordinals $\leq \varepsilon_0$

2.1 Informal Treatment of the Fast-Growing Hierarchy for ω

With the functions $f_m(n)$ we have so far, we can easily make enormous numbers like $f_{69}(420)$. But can we grow faster than that? If we *diagonlize* across all $f_m(n)$, we get $f_n(n)$ in red:

$\boldsymbol{f_0}$	$f_0(0)$	$f_0(1)$	$f_0(2)$	$f_0(3)$	$f_0(4)$
$\boldsymbol{f_1}$	$f_1(0)$	$f_1(1)$	$f_1(2)$	$f_1(3)$	$f_1(4)$
$\boldsymbol{f_2}$	$f_2(0)$	$f_2(1)$	$f_2(2)$	$f_2(3)$	$f_2(4)$
$\boldsymbol{f_3}$	$f_3(0)$	$f_3(1)$	$f_3(2)$	$f_3(3)$	$f_3(4)$
$\boldsymbol{f_4}$	$f_4(0)$	$f_4(1)$	$f_4(2)$	$f_4(3)$	$f_4(4)$

This $f_n(n)$ will eventually grow faster than any $f_m(n)$ like $f_{1000}(n)$. We can formalize this as such:

Definition 19

For 2 functions $f, g : \mathbb{N} \mapsto \mathbb{N}$, we say that f **dominates**, or **eventually dominates** g if there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $f(n) > g(n)$

We use $f_\omega(n)$ to denote $f_n(n)$. From this, we can continue counting up:
$f_{\omega+1}(n) = f_\omega^n(n)$
$f_{\omega+2}(n) = f_{\omega+1}^n(n)$
$f_{\omega 2}(n) = f_{\omega+n}(n)$
$f_{\omega 2+1}(n) = f_{\omega 2}^n(n)$
$f_{\omega^2}(n) = f_{\omega\omega}(n) = f_{\omega n}(n)$

One property of the fast-growing hierarchy is that it's *hard to go to the next tier*. For example, let's say we have a function that generates big numbers, $N(100) = f_{\omega^2}(100)$, and we want to try to "see" if we can reach $f_{\omega^3}(100)$ by using the $N()$ function. We can first try something like $N(100)^{N(100)}$, but that barely makes a dent googologoically. After all, exponentiation is at a measly f_2. What about nesting N, like $N(N(N(100)) = N^3(100)$? What if we go $N^{100}(100)$? Well, that is "just" $f_{\omega^2+1}(100) = f_{\omega^2}^{100}(100) = N^{100}(100)$. And to go to "just" $f_{\omega^2+2}(100)$, we have:

$$
\begin{aligned}
f_{\omega^2+2}(100) &= f^{100}_{\omega^2+1}(100) \\
&= f^{99}_{\omega^2+1}(f_{\omega^2+1}(100)) \\
&= f^{99}_{\omega^2+1}(N^{100}(100)) \\
&= f^{98}_{\omega^2+1}(f_{\omega^2+1}(N^{100}(100))) \\
&= f^{98}_{\omega^2+1}(N^{N^{100}(100)}(N^{100}(100)) \\
&> f^{98}_{\omega^2+1}\left(N^{N^{100}(100)}(100)\right) \\
&> f^{97}_{\omega^2+1}\left(N^{N^{N^{100}(100)}(100)}(100)\right)
\end{aligned}
$$

We can sort of see how $N^{N^{\cdot^{\cdot^{N^{100}(100)}}}(100)}(100)$ with 100 layers, is a lower bound of $f_{\omega^2+2}(100)$. Beyond that, it starts to get very tedious to even try to express $f_{\omega^2+3}(100)$, let alone $f_{\omega^2+\omega}(100)$. And ω^2 is honestly still quite a small ordinal! We can easily come up with a bigger ordinal to plug into the fast-growing hierarchy, like ω^ω. Our resulting number, $f_{\omega^\omega}(100)$, essentially cannot be expressed in terms of $N(100)$ at all (at least, not without using the fast-growing hierarchy).

2.2 Formal Treatment of the Fast-Growing Hierarchy for ordinals $\leq \varepsilon_0$

In the warning earlier, it specified that we need to specify a system of fundamental sequences for ordinals in order for FGH to be properly defined. Here, we will formally define fundamental sequences for ordinals $\leq \varepsilon_0$.

Before we define that, let's first have a standard way to express ordinals $\leq \varepsilon_0$:

Theorem 1 (**Cantor Normal Form**)

For every ordinal α there is a unique sequence of ordinals $\alpha_0 \geq \alpha_1 \geq \ldots \geq \alpha_n$ such that

$$\alpha = \omega^{\alpha_0} + \omega^{\alpha_1} + \ldots + \omega^{\alpha_n}$$

The basic rules of CNF is that you can only use addition and the "ω power" function $\omega^\bullet(\alpha) = \omega^\alpha$.

Note that what exactly the α_n in ω^{α_n} is written as is not enforced, so $\omega^{1+\omega}$ is technically a valid way to write ω^ω in CNF. Another way to illustrate it is that everything in **red** is enforced, while those in **black** are not:

$$
\begin{aligned}
\omega^\omega + \omega^2 \cdot 2 + \omega \cdot 3 &= \omega^\omega + \omega^2 + \omega^2 + \omega^1 + \omega^1 + \omega^1 \\
&= \omega^{1+\omega} + \omega^{1+1} + \omega^{\omega^0+\omega^0} + \omega^{\omega^0} + \omega^{S(0)} + \omega^{\text{the smallest ordinal greater than 0}}
\end{aligned}
$$

Now that we have a standardized way to uniquely write each ordinal, we can finally go into specifying fundamental sequences. The usual way to define these fundamental sequences based on Cantor Normal Form is the **Wainer Hierarchy**.

Definition 20

The **Wainer Hierarchy** is a system of fundamental sequences for ordinals $\leq \varepsilon_0$ (expressed in Cantor Normal Form):

1. $0[n] := 0$
2. $(\omega^{\alpha_0} + \omega^{\alpha_1} + \ldots + \omega^{\alpha_n})[n] := \omega^{\alpha_0} + \omega^{\alpha_1} + \ldots + \omega^{\alpha_n}[n]$ (i.e. take the fundamental sequence of the last term, which is the term with the smallest α)

3. $\omega^0[n] = 1[n] := 0$
4. $\omega^{\alpha+1}[n] = \omega^\alpha \cdot n$
5. $\omega^\alpha[n] = \omega^{\alpha[n]}$ if α is a limit ordinal
6. $\varepsilon_0[0] := 0$, and $\varepsilon_0[n+1] = \omega^{\varepsilon_0[n]}$

Example 12

If we want to find out what $f_{\omega^{\omega^2}+\omega^\omega}(3)$ is, we first evaluate $\left(\omega^{\omega^2}+\omega^\omega\right)[3]$

$$\begin{aligned}\left(\omega^{\omega^2}+\omega^\omega\right)[3] &= \omega^{\omega^2}+\omega^\omega[3]\\ &= \omega^{\omega^2}+\omega^{\omega[3]}\\ &= \omega^{\omega^2}+\omega^3\end{aligned}$$

So $f_{\omega^{\omega^2}+\omega^\omega}(3) = f_{\omega^{\omega^2}+\omega^3}(3)$, so we still need to evaluate $\left(\omega^{\omega^2}+\omega^3\right)[3]$:

$$\begin{aligned}\left(\omega^{\omega^2}+\omega^3\right)[3] &= \omega^{\omega^2}+\omega^2\cdot\omega[3]\\ &= \omega^{\omega^2}+\omega^2\cdot 3\end{aligned}$$

Then we find the fundamental sequence of *that*:

$$\begin{aligned}\omega^{\omega^2}+(\omega^2\cdot 3)[3] &= \omega^{\omega^2}+\omega^2\cdot 2+\omega^2[3]\\ &= \omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot\omega[3]\\ &= \omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 3\end{aligned}$$

And repeat again:

$$\begin{aligned}\omega^{\omega^2}+\omega^2\cdot 2+(\omega\cdot 3)[3] &= \omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+\omega[3]\\ &= \omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+3\end{aligned}$$

So $f_{\omega^{\omega^2}+\omega^\omega}(3) = f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+3}(3)$. Then

$$\begin{aligned}&f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+3}(3)\\ =\ &f^3_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}(3)\\ =\ &f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}(3)\right)\right)\\ =\ &f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f^3_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+1}(3)\right)\right)\\ =\ &f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f^{f^3_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+1}(3)}_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+1}\left(f^3_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+1}(3)\right)\right)\end{aligned}$$

And we will stop there. It's quite clear that this grows wayyy out of control that even expressing it in terms of $f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2}$ is basically impossible

Example 13

$$
\begin{aligned}
\varepsilon_0[0] &= 0 \\
\varepsilon_0[1] &= \omega^{\varepsilon_0[0]} = \omega^0 = 1 \\
\varepsilon_0[2] &= \omega^{\varepsilon_0[1]} = \omega^1 = \omega \\
\varepsilon_0[3] &= \omega^{\varepsilon_0[2]} = \omega^\omega \\
\varepsilon_0[4] &= \omega^{\varepsilon_0[3]} = \omega^{\omega^\omega} \\
&\dots
\end{aligned}
$$

So:

$$
\begin{aligned}
f_{\varepsilon_0}(1) &= f_1(1) \\
f_{\varepsilon_0}(2) &= f_\omega(2) \\
f_{\varepsilon_0}(3) &= f_{\omega^\omega}(3) \\
f_{\varepsilon_0}(4) &= f_{\omega^{\omega^\omega}}(4) \\
&\dots
\end{aligned}
$$

3 Other growing hierarchies

Aside from the fast-growing hierarchy, we have other slower hierarchies, though at sufficiently large ordinals they "catch up" with the fast-growing hierarchy.

Definition 21

Slow-growing hierarchy (SGH):
1. $g_0(n) := 0$
2. $g_{\alpha+1}(n) := g_\alpha(n) + 1$
3. $g_\alpha(n) = g_{\alpha[n]}$ when α is a limit ordinal

Definition 22

Middle-growing hierarchy (MGH):
1. $m_0(n) := n + 1$
2. $m_{\alpha+1}(n) := m_\alpha^2(n) = m_\alpha(m_\alpha(n))$
3. $m_\alpha(n) := m_{\alpha[n]}$ when α is a limit ordinal

Definition 23

Hardy hierarchy (HH):
1. $H_0(n) := n$
2. $H_{\alpha+1}(n) := H_\alpha(n+1)$
3. $H_\alpha(n) := H_{\alpha[n]}$ when α is a limit ordinal

05 Formal treatment of orders and ordinals

Order types are the most general way to define ordinals. Under this definition, an ordinal is an equivalence class of *well-ordered sets* under the relation of *order isomorphism*. There's a lot to unpack there, so we'll introduce the concepts one at a time:

1 Orders and well-orders

Definition 24

A **relation** R on a set X is a set of ordered pairs from X, i.e., $R \subseteq X \times X = \{(x, y) \mid x, y \in X\}$. We write xRy for $(x, y) \in R$

Specifically, such a relation is a *binary relation* or *2-ary relation* as it is a relation between 2 sets. However, we will mainly be using binary relations, so we will just call them relations.

Example 14

The relation "less than" on a set $\{1, 2, 3\}$ can be defined as the following set of ordered pairs from X: $R = \{(x, y) \mid x, y \in X \text{ and } x < y\} = \{(1, 2), (1, 3), (2, 3)\}$.

We see that R is a subset of $X \times X$:

$$\{(1, 2), (1, 3), (2, 3)\} \subset \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

Definition 25

A relation $<$ on X is called a **strict linear order** on X if and only if the following are true for all $x, y, z \in X$:

1. $<$ is transitive: If $x < y$ and $y < z$, then $x < z$
2. $<$ is asymmetric: If $x < y$, then $y \not< x$ (which is $\neg(y < x)$)
3. $<$ is total: If $x \neq y$, then either $x < y$ or $y < x$

For strict linear orders, we'll usually use symbol $<$ and its variants like $\prec$. The *partial linear order* $x \leq y$ means $x = y$ or $x < y$. The symbol $>$ can similarly be defined as $x > y$ if and only if $y < x$.

Definition 26

A relation $<$ is a **well-ordering** on the set X is a relation with a **strict linear ordering** of X such that **every (non-empty) subset of X has a least element under $<$**.

A least element of Y is some $y \in Y$ such that $y \leq z$ for every $z \in Y$, and is unique for a linearly ordered set (if a least element exists).

Definition 27

A **well-ordered set** $(X, <_X)$ is a set X and a relation $<_X$ that is a well-ordering of X

Proposition 2

Suppose $(X, <)$ is a well-ordered set. For every $x \in X$, if it is not the maximum element, then there is a unique $x' > x$ such that there is no $z \in X$ such that $x < z < x'$. (x' here is called the **successor** of x in $(X, <)$)

Proof

Let $Y = \{y \in X \mid y > x\}$, the set of all elements greater than x. Since $<$ is a well-ordering of X, then $Y \subseteq X$ must have a least element x'. Since $x' \in Y$, $x < x'$.

Since x' is the least element in Y, there can't be any $z \in Y$ such that $x < z$ and $z < x'$

To prove x' is unique, if we assuming there exists another number x'' with the same property that there is no z such that $x < z < x''$, we have to show $x' = x''$. Since $x < x''$, $x'' \in Y$. Since x' is the least element in Y, $x' \leq x''$. We can't have $x' < x''$, as it leads to $x < x' < x''$ which violates the property mentioned. Therefore $x' = x''$, and thus x' is unique.

Another property of well-ordered set is that they do not contain an infinite descending sequence. This property is needed to do induction along well-ordered sets. It also serves as another way to prove that an ordered set is well-ordered.

Proposition 3

Every well-ordered set does not contain an infinite descending sequence $x_1 > x_2 > \ldots$.

Similarly, every strictly linearly ordered set which does not contain an infinite descending sequence is a well-ordered set.

Proof

Suppose X is well-ordered by $<$ but yet there is an infinite descending sequence $x_1 > x_2 > \ldots$. We create a subset $Y \subseteq X$ such that $Y = \{x_i \mid i \in \mathbb{N} \text{ and } i \geq 1\}$. Since Y is a nonempty subset of X, it has a least element. But if $y \in Y$ is the least element, then such an infinite sequence would not be able to exist.

Suppose $(X, <)$ is a strictly linearly ordered set which does not contain an infinite descending sequence, but is not a well-ordered set. As it is not a well-ordered set, there exists a subset $P \subseteq X$ such that there is no least element. Therefore, for any $x_i \in P$, we can find an $x_{i+1} \in P$ such that $x_i > x_{i+1}$. We can construct an infinite descending sequence $x_1 > x_2 > x_3 > \ldots$, a contradiction. Therefore $(X, <)$ must be a well-ordered set.

We have a similar result for **non-increasing** sequences:

Proposition 4

If $(X, <)$ is a well-ordered set, and $x_1 \geq x_2 \geq x_3 \geq \ldots$ is a non-increasing infinite sequence, then the sequence eventually stabilizes, i.e. for some k, $x_k = x_i$ for all $i \geq k$.

Proof

Assume the converse, that for all k, $x_k \neq x_{i(k)}$ for some $i(k) > k$.

Since $x_k \geq \ldots \geq x_{i(k)}$, by transitivity, $x_k \geq x_{i(k)}$. But since $x_k \neq x_{i(k)}$, $x_k > x_{i(k)}$. This leads to an infinite descending sequence, which cannot exist in a well-ordered set, a contradiction.

2 Order Isomorphisms and Order Types

Definition 28

An **order isomorphism** between two *ordered* sets $(X, <_X), (Y, <_Y)$ is a bijection $f : S \mapsto T$ such that for all $x, y \in X$, $x <_X y \Leftrightarrow f(x) <_Y f(y)$.

If that was too dense, to put it in another way, it is a mapping $f : X \mapsto Y$ such that:
1. f is total, i.e., defined for all $x \in X$
2. f is injective, i.e., if $x \neq y$ then $f(x) \neq f(y)$
3. f is surjective onto Y, i.e. for every $y \in Y$, there is an $x \in X$ such that $f(x) = y$
4. f is order-preserving, i.e., $x <_X y$ if and only if $f(x) <_Y f(y)$

Note: the first three criteria are for the definition of a bijection.

If there is an order isomorphism between two well-ordered sets, we say they are **order isomorphic**, or that they have the same **order type**.

Note: an order isomorphism only requires that the sets be *ordered*. It does not require them to be *well-ordered*. Nonetheless we will usually be dealing with well-ordered sets, as the ordinals are well-ordered under $\in$:

Definition 29

Every ordinal α has the following properties:
1. transitivity, i.e., for all $\beta \in \alpha$, $\beta \subseteq \alpha$, and
2. $\in$ is a well-ordering on α

1. For all $\beta \in \alpha$, $\beta < \alpha$ and therefore the elements inside β would be inside α, i.e. $\beta \subseteq \alpha$.
2. An infinite descending sequence can't exist in α as it has a least element 0.

This is another way we can define von-Neumann ordinals, and leads to the same structure, where the successor of an ordinal α is $\alpha \cup \{\alpha\}$, and an ordinal is the set of all ordinals smaller than it.

Proposition 5

If $(\alpha_1, \alpha_2, \ldots)$ is a sequence of ordinals that is increasing (i.e., $\alpha_i < \alpha_{i+1}$) then:

$$\bigcup_i \alpha_i = \{x \mid x \in \alpha_i \text{ for some } i\}$$

is an ordinal, and is the *least ordinal* $\beta > \alpha_i$ for all $i = 1, 2, \ldots$.

The ordinal $\bigcup_i \alpha_i$ is known as the **limit**, **supremum**, or **strict supremum**[1] of a sequence α_i. We denote this with the the symbol $\overset{+}{\sup}$.

[1] In analysis, supremum usually uses partial $\geq$ rather than strict $>$: it usually refers to the least number s such that for all $x \in X$, $s \geq x$. Note that for limit ordinals, there is no difference between a strict supremum and the "typical" supremum.

Proof

Suppose $\alpha \in \bigcup_i \alpha_i$. Then for some $i, \alpha \in \alpha_i$. Since α_i is an ordinal, so are all of its elements, therefore α is also an ordinal. This means that every element in $\bigcup_i \alpha_i$ is an ordinal.

Now we have to show that $\bigcup_i \alpha_i$ contains every ordinal less than any member of it. This might sound confusing, but let's take a look at ω. ω contains every ordinal less than any member of it. For example, 500 is an ordinal in ω, and all ordinals < 500 are also contained inside ω. This is true for any ordinal $n \in \omega$.

So for any ordinal $\alpha \in \bigcup_i \alpha_i$, for some i, $\alpha \in \alpha_i$. Now consider all ordinals $\beta < \alpha$. The ordinal β has to be $\in \alpha_i$ as α_i is an ordinal and therefore contains all ordinals below it ($\beta < \alpha < \alpha_i$).

Therefore for any ordinal $\alpha \in \bigcup_i \alpha_i$, for all $\beta < \alpha$, $\beta \in \bigcup_i \alpha_i$, and as such $\bigcup_i \alpha_i$ is an ordinal.

Proposition 6

If $(X, <_X)$ and $(Y, <_Y)$ are order isomorphic and $(X, <_X)$ is a well-ordered set, $(Y, <_Y)$ is also a well-ordered set.

Proof

Let f be an order isomorphism between X and Y. Suppose$(X, <_X)$ is a well-ordered set but $(Y, <_Y)$ isn't. Then there is an infinite descending sequence $y_1 >_Y y_2 >_Y \ldots$. Because f is onto, for each i there exists there is an $x_i \in X$ such that $f(x_i) = y_i$. Since $y_i >_Y y_{i+1}, f(x_i) >_Y f(x_{i+1})$ and by the order-preserving property, $x_i >_X x_{i+1}$. But then $x_1 >_X x_2 >_X \ldots$ is an infinite descending sequence in X, a contradiction.

Example 15

Show that the set of natural numbers $\mathbb{N}$ and the set of even natural numbers $E = \{2n \mid n \in \mathbb{N}\}$, with their usual order, have the same order type.

$f(x) = 2x$ is an order isomorphism between $\mathbb{N}$ and E as:

1. f is defined for all $\mathbb{N}$
2. If $x \neq y, 2x \neq 2y$
3. For every $n \in E, n = 2 \cdot m$ where $m \in N$. m satisfies $f(m) = n$
4. If $x < y, 2x < 2y$ and vice versa

And in fact, since $\omega = \mathbb{N}$ is an ordinal, it is well-ordered. Therefore by this proposition, E which has an order isomorphism to ω is also well-ordered.

In the above example, we show that these two well-ordered sets are order isomorphic have the same **order type**. Since the set of all natural numbers is ω, we can generalize this to say that any set that has order isomorphism to $\omega = \mathbb{N}$ has order type ω. In general, **any set that has order isomorphism to an ordinal α has order type α**.

Example 16

Show that the set $X = \left\{\frac{1}{n} \mid n \in \mathbb{N} \text{ and } n \neq 0\right\} \cup \{0\}$ with the ordering $>$:

1. has the same cardinality as ω

2. is **not** order isomorphic to ω
3. is order isomorphic to $\omega + 1$
4. is a well-ordered set

(note that the ordering of X is "greater than" rather than the typical "lesser than". As such we will color $>$ to represent the ordering of X)

1. For cardinality, we do not need to consider the order of the set. We can define a bijection $f : \omega \mapsto X$ as such:

$$f(n) = \begin{cases} 0 & \text{if } n = 0 \\ \frac{1}{n} & \text{if } n \neq 0 \end{cases}$$

   (The validity of this bijection ~~should be obvious~~ is left as an exercise to the reader)

2. The main reason why X is not isomorphic to ω is that when you order the set by $>$, you get $\left\{\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \ldots, 0\right\}$. There is an element "at the end", whereas for $\omega = \{0, 1, 2, \ldots\}$ there is no such element at the end.

   Let's prove by contradiction. Suppose an order isomorphism $f : \omega \mapsto X$ exists. Then there must be some $n \in \omega$ that maps to 0, i.e., there must exist some n such that $f(n) = 0$.

   However, there exists an $n + 1 \in \omega$ such that $n < n + 1$. The order-preserving property therefore requires that $f(n) > f(n + 1)$, i.e., there must exist a $f(n + 1) \in X$ such that $0 > f(n + 1)$. But there is no such element in X, as $\frac{1}{n} > 0$ and $0 = 0$, a contradiction.

3. $\omega + 1 = \{0, 1, 2, \ldots, \omega\}$, so intuitively (though you shouldn't trust intuition when dealing with infinities) we can see how we can map this to X while preserving order.

   The following bijection $f : \omega + 1 \mapsto X$ is order isomorphic:

$$f(\alpha) = \begin{cases} \frac{1}{\alpha+1} & \text{if } \alpha < \omega \\ 0 & \text{if } \alpha = \omega \end{cases}$$

4. Since $\omega + 1$ is a well-ordered set, and there exists an order isomorphism from $\omega + 1$ to X, by this proposition X is a well-ordered set.

This is the crux of the difference between ordinals and cardinals. **Cardinality** is about comparing sizes via **bijections** between **unordered sets**, and assigning a **cardinal number**. **Order Types** is about comparing sizes via **order isomorphisms** between **(well-)ordered sets**, and assigning an **ordinal number**.

In our original definition, an ordinal is "an **equivalence class** of well-ordered sets under the relation of order isomorphism". We can interpret this as: "an ordinal represents the class of all well-ordered sets that are order isomorphic to each other". So in the equivalence class corresponding to $\omega + 1$, we have well-ordered sets like $(\{0, 1, 2, \ldots, \omega\}, <)$, and $\left(\left\{\frac{1}{n} \mid n \in \mathbb{N}, n \neq 0\right\} \cup \{0\}, >\right)$.

Now, let's look at how the concepts of well-orders and order types applies in other settings:

3 Lexicographical Ordering

Let us consider the ordering of the set of pairs of natural numbers $\mathbb{N}^2 = \{(n, m) \mid n, m \in \mathbb{N}\}$. The lexicographical ordering $\underset{\text{lex}}{\prec}$ is defined as:

$$(k, l) \underset{\text{lex}}{\prec} (n, m) \text{ if and only if (a) } k < n \text{ or (b) } k = n \text{ and } l < m$$

It's called a lexicographical ordering as it compares these pairs like how a dictionary would compare words - it compares the first element, and only moves onto the second element if the first element is the same.

$$(0,0) \underset{\text{lex}}{\prec} (0,1) \underset{\text{lex}}{\prec} (0,2) \underset{\text{lex}}{\prec} \dots \underset{\text{lex}}{\prec} (1,0) \underset{\text{lex}}{\prec} (1,1) \underset{\text{lex}}{\prec} \dots \underset{\text{lex}}{\prec} (2,0) \underset{\text{lex}}{\prec} \dots$$

We can extend this definition of lexicographical ordering to longer sequences. Similarly to $\mathbb{N}^2$, we call the set of all sequences of length k $\mathbb{N}^k$.

Definition 30

The **lexicographical ordering** $\underset{\text{lex}}{\prec}$ on $\mathbb{N}^k$ is defined as:

$(n_1, \dots, n_k) \underset{\text{lex}}{\prec} (m_1, \dots, m_k)$ if and only if, for some $j < k$, the following is true:
(a) $n_i = m_i$ for all i when $0 < i \leq j$
(b) $n_{j+1} < m_{j+1}$

Example 17

$(1,2,3) \underset{\text{lex}}{\prec} (3,2,1)$ because for $j = 0$:
(a) for all $0 < i \leq j, n_i = m_i$ (since there is no $0 < i \leq 0$, this is considered vacuously true)
(b) $n_1 < m_1$

$(1,2,3) \underset{\text{lex}}{\prec} (1,2,5)$ because for $j = 2$:
(a) for all $0 < i \leq j, n_i = m_i$
(b) $n_3 < m_3$

Personally, I find that lexicographical ordering defined via math is quite clunky. I think easier if you think about it conceptually (sorting it like how a dictionary would) than trying to find what j value satisfies the criteria.

With the definition of $\underset{\text{lex}}{\prec}$ extended to any sequence of length k, we can now show that:

Proposition 7

$\left(\mathbb{N}^k, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Proof

Let $Y \subseteq \mathbb{N}^k$. We now have to prove there exists a least element in Y.

Let n_1 be the smallest n such that $(n, m_2, m_3, \dots, m_k) \in Y$. Then there is no $(m_1, m_2, m_3, \dots, m_k) \in Y$ such that $m_1 < n_1$. Now let n_2 be the smallest n such that $(n_1, n, m_3, \dots, m_k) \in Y$. Again, there is no $(m_1, m_2, m_3, \dots, m_k) \in Y$ such that $m_1 < n_1$ or $n_1 = m_1$ and $m_2 < n_2$. Continue this process until you finally define n_k as the least n such that $(n_1, n_2, n_3, \dots, n) \in Y$. Then $(n_1, n_2, n_3, \dots, n_k) \in Y$ is the least element in the $\underset{\text{lex}}{\prec}$ ordering.

However, what if we want to compare sequences of arbitrary length? Let $\mathbb{N}^*$ be the sequences of natural number of arbitrary length:

$$(0), (1), (2), \ldots$$
$$(0,0), (0,1), (0,2), \ldots$$
$$(0,0,0), (0,0,1), (0,0,2), \ldots$$

We can extend our usual definition of lexicographic ordering as such:

Definition 31

For 2 sequences s, t:

$$s = (n_1, n_2, \ldots, n_k)$$
$$t = (m_1, m_2, \ldots, m_l)$$

$s \underset{\text{lex}}{\prec} t$ if and only if either of the following is true:

1. $n_1 = m_1, n_2 = m_2, \ldots, n_k = m_k$, and $k < l$

   Example 18

   $(1,3,5) \underset{\text{lex}}{\prec} (1,3,5,7,9)$

2. there is a $j < k$ such that $n_1 = m_1, \ldots, n_j = m_j$ and $n_{j+1} < m_{j+1}$

   Example 19

   $(1,3,5,7,9) \underset{\text{lex}}{\prec} (1,3,7)$

   The first two elements are the same but in the third element they are different. The third element in the first sequence (5) < the third element in the second sequence (7).

   Put another way, $j = 2, n_1 = m_1, n_2 = m_2, n_3 < m_3$

However, $\left(\mathbb{N}^*, \underset{\text{lex}}{\prec}\right)$ is **not a well-ordered set**, as we can construct an infinite descending sequence of elements:

$$(1) \underset{\text{lex}}{\succ} (0,1) \underset{\text{lex}}{\succ} (0,0,1) \underset{\text{lex}}{\succ} \ldots$$

However, if we only consider **decreasing sequences**, $\underset{\text{lex}}{\prec}$ is a well-order (Note that in the above example, none of them are decreasing.)

Proposition 8

Let $\mathbb{N}^*_>$ be the set of all decreasing sequences $(n_1, n_2, \ldots, n_k)$ where $n_1 > n_2 > \ldots > n_k$.

$\left(\mathbb{N}^*_>, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Proof

Suppose for contradiction that there exists an infinite descending sequence of elements in $\mathbb{N}^*_>$:

$s_1 \underset{\text{lex}}{\succ} s_2 \underset{\text{lex}}{\succ} \ldots$, where $s_1, s_2, \ldots \in \mathbb{N}^*_>$.

Let $s_i = \left(n_1^i, n_2^i, \ldots, n_{k_i}^i\right) \in \mathbb{N}^*_>$.

If we consider the first elements of $s_1, s_2, s_3, \ldots$ we get a non-increasing sequence $n_1^1 \geq n_1^2 \geq n_1^3 \geq \ldots$. For every k, $n_1^k \geq n_1^{k+1}$, as if $n_1^k < n_1^{k+1}$, then lexicographically $s_k \underset{\text{lex}}{\prec} s_{k+1}$ and it won't be in the infnite descending sequence.

Since it is a non-increasing sequence in the well-ordered set $\mathbb{N}$, it must stabilize at some index. Let's call the index j_1, and the value it stabilizes at m_1. Therefore for all $i \geq j_1$, the first element of s_i must be m_1: $s_i = \left(m_1, n_2^i, n_3^i, \ldots, n_{k_i}^i\right)$.

Now let's consider all sequences after the index j_1. We can similarly construct a non-increasing sequence from the second elements in those sequences: $n_2^{j_1} \geq n_2^{j_1+1} \geq n_2^{j_1+2} \geq \ldots$. It similarly stabilizes at an index j_2 with a value of m_2. Therefore for all $i \geq j_2$, the second element of s_i must be m_2: $s_i = (m_1, m_2, n_3^i, \ldots, n_{k_i}^i)$.

Now since $s_i \in \mathbb{N}_<^*$, s_i *is a decreasing sequence*, so $m_1 > m_2$. Continuing this, we get $m_1 > m_2 > m_3 > \ldots$, which is an infinite descending sequence in $\mathbb{N}$, a contradiction. Therefore $\left(\mathbb{N}_>^*, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Note that there's nothing special about $\mathbb{N}$ here, we can replace $\mathbb{N}$ with an well-ordered set:

Proposition 9

Let $(X, <)$ be a well-ordered set. Let $X_>^*$ be the set of all decreasing sequences $(x_1, x_2, \ldots, x_k)$ where $x_1 > x_2 > \ldots > x_k$ and $x_i \in X$.

$\left(X_>^*, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

In fact, the set of all **non-increasing** sequences of elements of a well-ordered set X is also a well-ordered by $\underset{\text{lex}}{\prec}$. A non-increasing sequence of elements in X is a sequence $(x_1, x_2, \ldots, x_k)$ where $x_i \geq x_{i+1}$ (rather than strictly decreasing $x_i > x_{i+1}$).

To show this, we can split non-increasing sequences into a sequence of constant sequences, e.g. $(4, 4, 4, 3, 2, 2, 1, 1, 1) \to ((4, 4, 4), (3), (2, 2), (1, 1, 1, 1))$.

Lemma 1

Let $(X, <)$ be a well-ordered set. Let $X_=^*$ be the set of all constant sequences $(x, \ldots, x) \in X$.

e.g. if $X = \{1, 2, 3\}$, $X_=^* = \{(1), (1, 1), (1, 1, 1), \ldots, (2), (2, 2), \ldots, (3), (3, 3), \ldots\}$

$\left(X_=^*, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Proof

Let $Y \subseteq X_=^*$. Now we have to show there exists a least element in Y.

Lexicographically, we compare sequences by their first entry. As such, the least element must have the least first entry. Let x be the value of the least first entry.

Since all sequences in $X_=^*$ are constant sequences, all candidates to be the least element will be sequences like $(x, x), (x, x, x, x), (x, x, x)$. The least element (lexicographically) would be the *shortest* constant sequence of x.

Proposition 10

If $(X, <)$ is a well-ordered set, the set of non-increasing sequences of elements $\left(X^*_{\geq}, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Proof

Since the set of constant sequences $X^*_{=}$ is well-ordered, by this proposition, the set of decreasing sequences of constant sequences $(X^*_{=})^*_{\underset{\text{lex}}{\succ}}$ is also well-ordered. Now, by this proposition all we need to do is construct an order ismorphism from $(X^*_{=})^*_{\underset{\text{lex}}{\succ}}$ to $X^*_{\geq}$ to prove that $X^*_{\geq}$ is a well-ordered set.

Let y_i denote constant sequences in $X^*_{=}$. We can construct an isomorphism $f : (X^*_{=})^*_{\underset{\text{lex}}{\succ}} \mapsto X^*_{\geq}$:

$$f((y_1, y_2, ..., y_{k'})) = \left(\underbrace{x_1, ..., x_1}_{\text{contents of } y_1}, \underbrace{x_2, ..., x_2}_{\text{contents of } y_2}, ..., \underbrace{x_{k'}, ..., x_{k'}}_{\text{contents of } y_{k'}}\right)$$

Both sets are ordered lexicographically. $X^*_{\geq}$ is ordered by $\underset{\text{lex}}{\prec}$, and $(X^*_{=})^*_{\underset{\text{lex}}{\succ}}$ is ordered lexicographically based on $\underset{\text{lex}}{\prec}$, so this is an order-preserving isomorphism.

Therefore is $X^*_{\geq}$ is a well-ordered set.

All this discussion on lexicographic orderings is quite abstract, but we will use the concepts of orders (especially on non-increasing sets) in the next chapter.

06 Ordinal Notations (up to ε_0)

An **ordinal notation**, or **ordinal notation system** is basically a way to express an ordinal as a **finite string**, i.e., finite sequences of finite symbols. Think of the decimal system, that allows you to express any number into a finite sequence of the symbols $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$. Of course, we also need to define some ordering of these strings too.

Definition 32

An **Ordinal Notation** is a well-ordered set $(OT, \prec)$, where OT is a *recursive* set of finite formal strings with a finite alphabet, and $\prec$ is a *recursive* order.

(*recursive* here means that there exists an algorithm to compute them, i.e. computable)

1 Iterated Cantor Normal Form

Definition 33

Iterated Cantor Normal Form (ICNF) is an ordinal notation for ordinals $< \varepsilon_0$.

The ordinal notation here is a string consisting of finitely many of the following symbols: (Only the symbols highlighted in red are allowed to be used in ICNF.)

- 0
- $+$
- The function $\omega^{\bullet}(\text{something})$ which can also be written as $\omega^{\text{something}}$

The ICNF ordinal notation $o(\alpha)$ for an ordinal $\alpha < \varepsilon_0$ is:

$$o(\alpha) := \begin{cases} 0 & \text{if } \alpha = 0 \\ \omega^{o(\alpha_1)} + \omega^{o(\alpha_2)} + \ldots + \omega^{o(\alpha_n)} & \text{if the CNF of } \alpha \text{ is } \omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n} \end{cases}$$

This works as a notation because Cantor normal form is unique, and for $\alpha < \varepsilon_0$, $\alpha_i < \alpha$.

We can define a well-ordering $\prec$ that borrows from the normal order of ordinals by saying $o(\alpha) \prec o(\beta)$ if and only if $\alpha < \beta$. However, such an ordering is not recursive as showing that $\alpha \in \beta$ requires using infinite sets. We can order the strings on our own without relying on the order of ordinals (we will define one later).

Example 20

$$\begin{aligned} o(\omega^\omega + \omega^3 + \omega^2 + 1) &= o(\omega^\omega + \omega^3 + \omega^2 + \omega^0) \text{ (convert to CNF within } o()) \\ &= \omega^{o(\omega)} + \omega^3 + \omega^2 + \omega^0 \\ &= \omega^{o(\omega^1)} + \omega^{o(\omega^0+\omega^0+\omega^0)} + \omega^{o(\omega^0+\omega^0)} + \omega^{o(\omega^0)} \\ &\text{(convert to CNF within } o()) \\ &= \omega^{\omega^{o(1)}} + \omega^{\omega^0+\omega^0+\omega^0} + \omega^{\omega^0+\omega^0} + \omega^{\omega^0} \\ &= \omega^{\omega^{\omega^0}} + \omega^{\omega^0+\omega^0+\omega^0} + \omega^{\omega^0+\omega^0} + \omega^{\omega^0} \\ &\text{(convert to CNF within } o()) \end{aligned}$$

Definition 34

Here we will define a recursive ordering for the ICNF ordinal notation. We define the following inductively:

1. the set $OT_{=k}$ of ordinal notations of height k,
2. the set $OT_{\leq k}$ of ordinal notations of height $\leq k$,
3. an ordering $\prec_k$ on $O_{\leq k}$

The sets $OT_{=k}$ and $OT_{\leq k}$ are defined as such:

1. $\mathbf{0}$ is the only ordinal notation of height 0, so $OT_{=0} = OT_{\leq 0} = \{\mathbf{0}\}$, and $\prec_0 = \{\}$, i.e., $\mathbf{0} \not\prec_0 \mathbf{0}$
2. If $OT_{=k}, OT_{\leq k}, \prec_k$ are defined, then the set $OT_{=k+1}$, the set of all ordinal notations of height $k+1$, is defined as:

$$\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n}$$

where $\alpha_1 \in OT_{=k}$, and $\alpha_1 \succeq_k \alpha_2 \succeq_k \ldots \succeq_k \alpha_n$ ($\alpha_i \in OT_{\leq k}$).

We color greek symbols in **green** to represent ordinal notations, rather than ordinals. Note that those colored in **red** still represents the symbols themselves. To use a programming analogy, symbols in **green** are like variables containing strings (specifically, ordinal notation strings).

The order $\prec_{k+1}$ on $OT_{<k+1}$ is defined as such:

- Case 1: α and β are both of height $\leq k$ (α and $\beta \in OT_{\leq k}$).

  In this case, we just use $\prec_k$ to compare them, i.e. $\alpha \prec_{k+1} \beta$ if $\alpha \prec_k \beta$.

- Case 2: α is of height $\leq k$ but β is of height $k+1$ ($\alpha \in OT_{\leq k}$ and $\beta \in OT_{=k+1}$). In this case, $\alpha \prec_{k+1} \beta$

- Case 3: α and β are both of height $k+1$.

  In this case, we can express α and β as such they are of the form:

$$\alpha = \omega^{\alpha_1} + \ldots + \omega^{\alpha_n}$$
$$\beta = \omega^{\beta_1} + \ldots + \omega^{\beta_m}$$

  We order this lexicographically, comparing α_1 and β_1. If they are equal, move onto α_2 and β_2, and if they are still equal, move on until either you find a $\alpha_i \prec \beta_i$ or, if all exponents are equal, that α has less terms than β. A formal way to write this is:

  $\alpha \prec_{k+1} \beta$ if either of the following hold:

  - $\alpha_i = \beta_i$ for all $i \leq j$ and $j = n < m$

    e.g. $\omega^{\omega^0} + \omega^0 \prec_2 \omega^{\omega^0} + \omega^0 + \omega^0$, here $n = 2, m = 3, j = n = 2$.

  - $\alpha_i = \beta_i$ for all $i \leq j$ and $\alpha_{j+1} < \beta_{j+1}$, $j < n$

    e.g. $\omega^{\omega^0} + \omega^0 \prec_2 \omega^{\omega^0} + \omega^{\omega^0}$, here $j = 1$

Finally, we can define the set of all ordinal notations $OT = OT_{\leq 0} \cup OT_{\leq 1} \cup OT_{\leq 2} \cup \ldots$, and an ordering $\prec = \prec_0 \cup \prec_1 \cup \prec_2 \cup \ldots$

We still have to prove that ordering $\prec$ on the set OT is a well-ordering. We will assume that it is a strict linear ordering – the proof of that is left as an exercise to the reader (totally not because I'm lazy).

Proposition 11

$(OT_{\leq k}, \prec)$ is a well-ordered set

Proof

$OT_{\leq 0} = \{0\}$ is trivially well-ordered, any nonempty subset is just $\{0\}$, of which 0 is the smallest element.

Now for induction, assume $(OT_{\leq k}, \prec)$ is well-ordered. The set $(OT_{\leq k+1}, \prec_{k+1})$ is order isomorphic to the set of nonincreasing sequences of ordinal notations from $(OT_{\leq k}, \prec_k)$. Why? because we can make a bijection $f : OT_{\leq k+1} \mapsto OT^*_{\leq k}$ like this:

$f(\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n}) = (\alpha_1, \alpha_2, \ldots, \alpha_n)$.

It should be apparent that the expression in the left hand side $\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n} \in OT_{\leq k+1}$, and $\alpha_1 \succeq \alpha_2 \succeq \ldots \succeq \alpha_n$, and thus the right hand side is an non-increasing sequence of expressions $\alpha_i \in OT_{\leq k}$ of variable length.

Since $OT_{\leq k+1}$ is order isomorphic to the set of all non-increasing sequence of elements from $OT_{\leq k}$, and $OT_{\leq k}$ is well-ordered, by this proposition, $(OT_{\leq k+1}, \prec)$ must therefore be well-ordered too.

Proposition 12

$(OT, \prec)$ is well-ordered

Proof

Let $\alpha_1 \succ \alpha_2 \succ \ldots$ be an infinite descending sequence of ordinal notations. The first α_1 is of some height k. Since any ordinal notation of height $> k$ would be greater than any ordinal notation of height k, the height of every α_i is $\leq k$. So $\alpha_1 \succ \alpha_2 \succ \ldots$ is an infinite descending sequence in $(OT_{\leq k}, \prec_k)$. However, $(OT_{\leq k}, \prec_k)$ is well-ordered, a contradiction. Therefore no infinite descending sequence can exist in $(OT, \prec)$, so it is a well-ordered set.

While ordinal notations look like actual ordinals with symbols like ω^0, they are really just complicated sequences of sequences of ... sequences of symbols. Ordinal notations of height 0 is just 0. Then the ordinal notations of height 1 are ω^0, $\omega^0 + \omega^0$, $\omega^0 + \omega^0 + \omega^0$, ... But we can instead represent them as sequences of ordinal notations of height 0, so $\omega^0 = (0)$, $\omega^0 + \omega^0 = (0, 0)$, and so on. We can then lexicographically order these sequences of 0 by length to get our order for $OT_{\leq 1}$, letting 0 be smaller than any sequence of 0s:

$$0 \prec_1 (0) \prec_1 (0, 0) \prec_1 (0, 0, 0) \prec_1 \ldots$$

With this order, we can make a set of non-decreasing sequences of terms from $OT_{\leq 1}$, which corresponds to $OT_{\leq 2}$. For example, $\omega^{\omega^0 + \omega^0} + \omega^0 = ((0, 0), 0)$. We can define a lexicographic order on these sequences based on the order $\prec_1$ to get $\prec_2$, and with this method we can keep building ordinal notations of greater and greater height.

Ordinal notations themselves are **not transfinite sets** (unlike actual ordinals) – this is why they are highlighted so distinctly in red. If you look closely, we do not need to know anything about transfinite ordinals to show that they are well-ordered (we just needed to know facts about lexicographical orderings of sequences).

2 Ordering of Trees and Goodstein sequences

We now have two order-isomorphic well-ordered sets of order type ε_0:

1. The ordinals $< \varepsilon_0$ ordered by $<$, or equivalently $\in$
2. Ordinal notations $< \varepsilon_0$ ordered by $\prec$ (This ordinal notation is known as Iterated Cantor Normal Form)

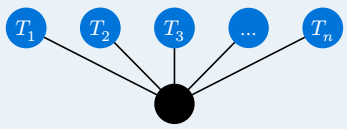
Another example of a well-ordered set of order type ε_0 is the set of all finite, finitely branching trees.

Definition 35

The set T of finite, finitely branching tree can be inductively defined as such:

1. The tree containing a single node is a tree of height 1. We can represent this as $()$ or as a diagram:

2. If $T_1, ..., T_n$ are trees, and k is the maximum height among $T_1, ..., T_n$, a tree of height $k+1$ is $(T_1, T_2, ..., T_n)$

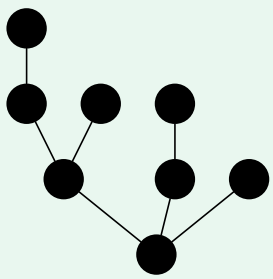


(nodes in blue represents trees)

3. Nothing else is a tree

Example 21

The tree `((((),()),()),(()),())` is of height 4:

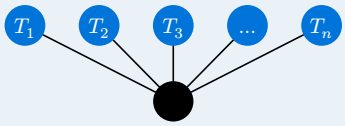


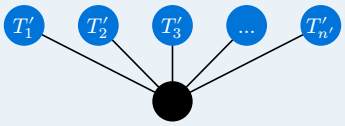
Definition 36

Now let's inductively define an order $<$ on these trees:

1. The tree of height 1 is only equal to itself, and is $<$ all trees of height > 1
2. Suppose T and T' are trees of height > 1, and let h be the maximum of their heights. Assume that $<$ has been defined for trees of height $< h$.

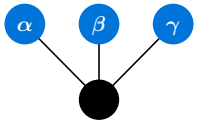
We can then write $T = (T_1, T_2, ..., T_n)$ and $T' = (T'_1, T'_2, ..., T'_{n'})$, where $T_i \geq T_{i+1}$, $T'_i \geq T'_{i+1}$ (i.e. arranged in a non-decreasing sequence)

$T =$ 

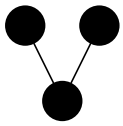
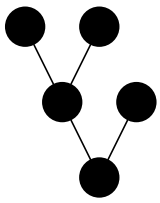
$T' =$ 

We can then order these lexicographically, i.e. $T < T'$ if:
1. $T_i = T'_i$ and $n < n'$
2. There exists a k such that for all $i < k$, $T_i = T'_i$ and $T_k < T'_k$

This should already be sounding familiar to you – this is analogous to our ordinal notation system, where we can correspond $\omega^\alpha + \omega^\beta + \omega^\gamma$ to a tree like this:



Some examples are

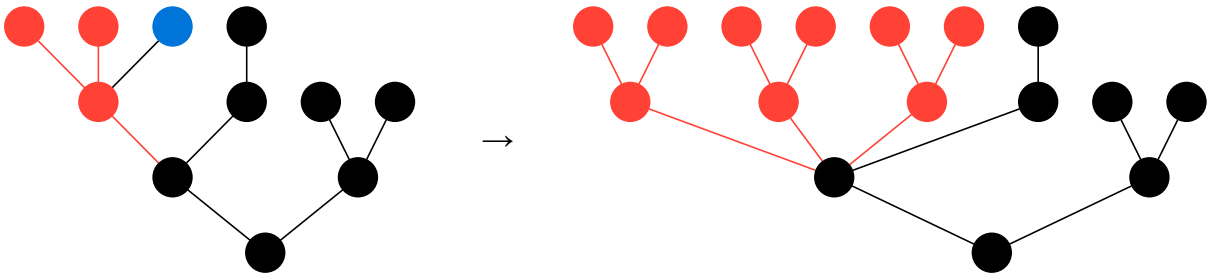
Ordinal notation	Associated tree
0	
ω^0	
$\omega^0 + \omega^0$	
$\omega^{\omega^0+\omega^0} + \omega^0$	

With this, we can construct an order isomorphism from T to ε_0, and show that T is of order type ε_0. We can basically show via induction that $OT_{\leq k}$, the set of all ordinal notations of height $\leq k$ is order isomorphic to $T_{\leq k}$, the set of all trees of height $\leq k$ since the way the ordering is defined (lexicographic ordering of elements of lower height) is the same. (informal proof)

2.1 Kirby-Paris Hydra

Imagine the previous tree, where every leaf node (i.e. not connected to any node above) is a head of a Hydra. Every time we cut off a head, the leaf's parent node and all of its remaining descendents will be multiplied n times.

Here's is an example: if we cut the blue leaf node, the part in red gets duplicated $n = 3$ times:



Theorem 2 (**Kirby-Paris Hydra**)

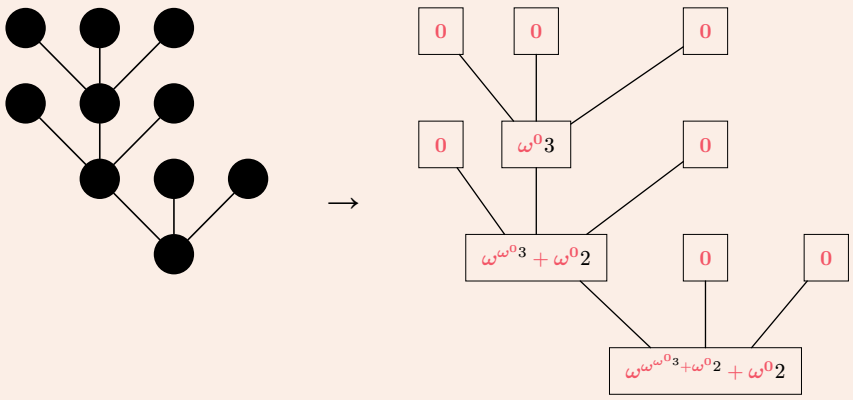
If you keep cutting off heads (in any manner), the Hydra eventually dies.

Proof

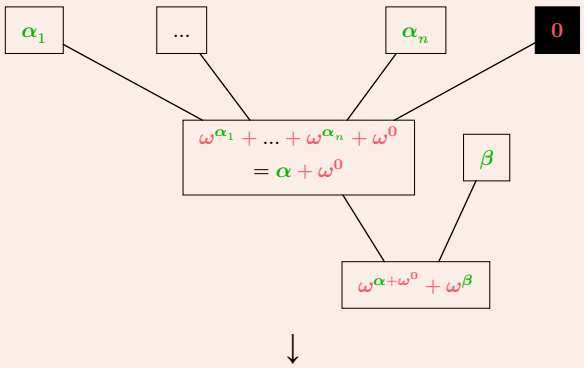
We assign ordinal notations $< \varepsilon_0$ to nodes in the tree:

1. Leaf nodes are assigned 0.
2. If the ordinal notations assigned to a node's descendants are $\alpha_1 \succeq \alpha_2 \succeq \ldots \succeq \alpha_n$, the node itself is assigned $\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n}$

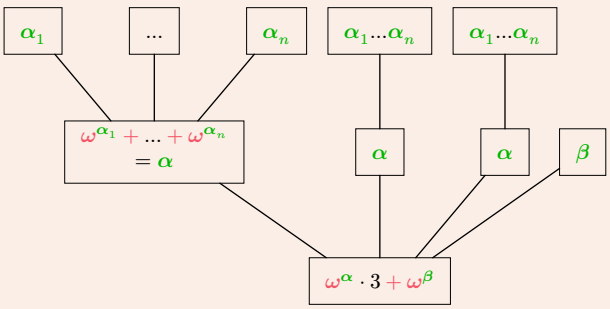
For example:



Now let's look at what happens when a leaf node (filled in black) is removed. We set $n = 3$ for this example. Here we denote $\alpha = \omega^{\alpha_1} + \ldots \omega^{\alpha_n}$



$\downarrow$



The ordinal notation at the bottom decreases whenever we remove a head:

$$\omega^{\alpha} \cdot n < \omega^{\alpha+\omega^0} \text{ since}$$
$$\omega^{\alpha} \cdot n < \omega^{\alpha+1}$$

Therefore, eventually, the root will reach 0 and the hydra will die.

2.2 Goodstein sequences

In base 10, we represent a number using powers of 10:

$$25739 = 2 \cdot 10^4 + 5 \cdot 10^3 + 7 \cdot 10^2 + 3 \cdot 10^1 + 9$$

In other bases, e.g. base 3, we can do the same:

$$25739 = 3^9 + 2 \cdot 3^7 + 2 \cdot 3^4 + 2 \cdot 3^3 + 2 \cdot 3 + 2$$

We can take a page out of our Iterated Cantor Normal Form and demand that the exponents of 3 must also be in terms of power of 3:

$$25739 = 3^{3^2} + 2 \cdot 3^{2 \cdot 3+1} + 2 \cdot 3^{3+1} + 2 \cdot 3^3 + 2 \cdot 3 + 2$$

We call this "hereditary" base-3 representation.

Now the next term in the **Goodstein sequence** for this number would be equivalent to taking every "3^" and replacing it with a "4^", then subtracting 1 from it.

$$4297067018 = 4^{4^2} + 2 \cdot 4^{2 \cdot 4+1} + 2 \cdot 4^{4+1} + 2 \cdot 4^4 + 2 \cdot 4 + 2 - 1$$

Definition 37

The **Goodstein sequence** for some number n is the sequence $(n_2, n_3, n_4, ...)$ where the number n_{i+1} is obtained by writing n_i in hereditary base-i notation, changing the is into $(i+1)$s and subtracting 1.

Example 22

Let's start with $n_2 = 10$. We first write 10 in hereditary base-2 notation:

$$n_2 = 2^{2+1} + 2$$

Then n_3 is:

$$n_3 = 3^{3+1} + 3 - 1 = 3^{3+1} + 2 = 83$$

We can repeat this for $n_4, n_5, \ldots$:

$$\begin{aligned}
n_4 &= 4^{4+1} + 2 - 1 = 4^{4+1} + 1 = 1025 \\
n_5 &= 5^{5+1} + 1 - 1 = 5^{5+1} = 15625 \\
n_6 &= 6^{6+1} - 1 = 279935 \\
&= 5 \cdot 6^6 + 5 \cdot 6^5 + 5 \cdot 6^4 + 5 \cdot 6^3 + 5 \cdot 6^2 + 5 \cdot 6 + 5 \\
n_7 &= 5 \cdot 7^7 + 5 \cdot 7^5 + 5 \cdot 7^4 + 5 \cdot 7^3 + 5 \cdot 7^2 + 5 \cdot 7 + 5 - 1 \\
&= 5 \cdot 7^7 + 5 \cdot 7^5 + 5 \cdot 7^4 + 5 \cdot 7^3 + 5 \cdot 7^2 + 5 \cdot 7 + 4 \\
&= 4215754
\end{aligned}$$

The goodstein sequence seems to grow forever, but this sequence eventually terminates at 0!

Theorem 3

The Goodstein sequence for n eventually reaches 0, for all n

Proof

let $\alpha(n, b)$ be the ordinal notation resulting from it by replacing every b by ω.

With this, for each n_i in the Goodstein sequence, we can assign an ordinal notation $\alpha(n_i, i)$ to it. We now show that $\alpha(n_{i+1}, i+1) \prec \alpha(n_i, i)$.

Suppose $n_i = a_k i^k + \ldots + a_j i^j + a_0$. Then $\alpha(n_i, i) = \omega^k \cdot a_k + \ldots + \omega^j \cdot a_j + a_0$.

If $a_0 > 0$, then

$$\begin{aligned}
n_{i+1} &= a_k (i+1)^k + \ldots + a_j (i+1)^j + a_0 - 1 \\
\alpha(n_{i+1}, i+1) &= \omega^k \cdot a_i + \ldots + \omega^j \cdot a_j + (a_0 - 1) \\
&\prec \omega^k \cdot a_k + \ldots + \omega^j \cdot a_j + a_0 = \alpha(n_i, i)
\end{aligned}$$

If $a_0 = 0$, then:

$$\begin{aligned}
n_{i+1} &= a_k (i+1)^k + \ldots + a_j (i+1)^j - 1 \\
&= a_k (i+1)^k + \ldots + (a_j - 1)(i+1)^j + (i+1)^j - 1 \\
&= a_k (i+1)^k + \ldots + (a_j - 1)(i+1)^j + \sum_{s=0}^{j-1} i(i+1)^s \\
\alpha(n_{i+1}, i+1) &= \omega^k \cdot a_k + \ldots + \omega^j \cdot (a_j - 1) + \sum_{s=0}^{j-1} \omega^s \cdot i \\
&\prec \omega^k \cdot a_k + \ldots + \omega^j \cdot a_j + a_0 = \alpha(n_i, i)
\end{aligned}$$

In the last line, we compare the factor on ω^j (the largest power of ω in which they differed), and find that $\alpha(n_i, i) \succ \alpha(n_{i+1}, i+1)$ no matter the value of a_0, and therefore the sequence reaches 0.

3 The Primitive Sequence System (PrSS)

Definition 38 (**Primitive Sequence System (PrSS)**)

Let the set T be the set of sequences $s = (a_1, a_2, ..., a_l)$ such that:

1. s is either empty $()$ or starts with 0: $(0, ...)$.
2. For any a_i in s, let it's **parent** a_j be the *latest element* before a_i such that $a_j < a_i$. a_j must be equal to $a_i - 1$.

   Put another way, let the *parent index* $p(i) \in \mathbb{N}$ be the largest possible $p(i) < i$ such that $a_{p(i)} < a_i$. For all a_i in the sequence, $a_{p(i)} = a_i - 1$.

   Example 23

   $(0, 1, 3)$ is not a valid sequence as the latest element before 3 that is less than 3 is 1, which is not $3 - 1 = 2$.

   $(0, 1, 2, 1, 2)$ is a valid sequence:
   - It starts with 0
   - $a_2 = 1$. The latest element less than 1 is $a_1 = 0$, which is 1 less than $a_2 = 1$
   - $a_3 = 2$. Similarly, the latest element less than 2 is $a_2 = 1$ which is 1 less than $a_3 = 2$
   - $a_4 = 1$. The latest element less than $a_4 = 1$ is $a_1 = 0$, so no issues
   - $a_5 = 2$. The latest element less than $a_5 = 2$ is $a_4 = 1$, so no issues

3. The ordering $<$ is simply the lexicographic ordering of these sequences.

We can use this notation to define either *a fast-growing function*, or a *system of fundamental sequences*:

Definition 39

For a sequence $s = (a_1, a_2, ..., a_l) \in T$, we define $s[n]$ as the nth member in its *fundamental sequence*.
- $()[n] = n$. Otherwise, for nonempty s:
- The **good root** g and the **bad root** b are defined as such:
  - $g = \left(a_1, ..., a_{p(l)-1}\right)$
  - $b = \left(a_{p(l)}, ..., a_{l-1}\right)$
  - if $p(l)$ does not exist (i.e. there is no a_i < a_l), then: $g = (a_1, ..., a_{l-1})$ and b is empty
- $s[n] = (g, b, b, ..., b)$ with n copies of b, though this can be replaced with any increasing function $f(n)$. The original definition used $f(n) = n^2$.

As for the *fast-growing function* also confusingly labelled $s[n]$:
- We split the sequence into good root g and bad root b again.
- $s[n] = (g, b, b, ..., b)[f(n)]$ with $f(n)$ copies of b. The original definition used $f(n) = n^2$, but $f(n) = n + 1$ can also be used.

The final result is the value in the [] when the sequence is eventually empty.

Example 24

Expand the following:

1. $(0, 0)[3]$
2. $(0, 1)[3]$
3. $(0, 1, 2, 3, 2)[3]$
4. $(0, 1, 2, 3, 4, 3, 4, 2, 3)[3]$

Seriously, try it yourself. I'll provide the answers in case you get stuck:

1. 0 has no parent, so good root is 0 and bad root is empty. We get (0). Note that n doesn't matter when the last element is 0.
2. The parent of 1 is 0, so we have an empty good root and a bad root of 0. We replicate the bad root $n = 3$ times, so we get $(0, 0, 0)$
3. The parent of 2 is 1, so we have a good root of 0 and a bad root of $1, 2, 3$, so we get $(0, 1, 2, 3, 1, 2, 3, 1, 2, 3)$
4. $(0, 1, 2, 3, 4, 2, 3, 2, 2, 2)$

Example 25

Evaluate $(0, 1, 1)[1]$ as a fast-growing function with $f(n) = n + 1$

$$\begin{aligned}
&(0, 1, 1)[1] \\
&= (0, 1, 0, 1)[2] \\
&= (0, 1, 0, 0, 0)[3] \\
&= (0, 1, 0, 0)[4] \\
&= (0, 1, 0)[5] \\
&= (0, 1)[6]
\end{aligned}$$

At this point we see a new pattern: $(a_1, ..., a_{l-1}, 0)[n] = (a_1, ..., a_{l-1})[f(n)]$, since there's no parent for 0 as there are no $a_i < 0$.

$$\begin{aligned}
&(0, 1)[6] \\
&= (0, 0, 0, 0, 0, 0, 0)[7] \\
&= (0, 0, 0, 0, 0, 0)[8] \\
&= (0, 0, 0, 0, 0)[9] \\
&= (0, 0, 0, 0)[10] \\
&= (0, 0, 0)[11] \\
&= (0, 0)[12] \\
&= (0)[13] \\
&= ()[14] \\
&= 14
\end{aligned}$$

Now imagine if we used a bigger value of n, making many more copies of the bad root. Or image if we used $f(n) = n^2$ instead of $n + 1$. $s[n]$ is a very powerful function. But we can constrain it's growth rate in the Hardy hierarchy as we'll show later.

3.1 Order isomorphism to ε_0

We now show that PrSS is order isomorphic to ε_0.

Now, just like in ICNF where we use a non-increasing sequence of ordinal notations of height $\leq k$ to define a new height $k+1$:

$$\alpha_1 \geq \alpha_2 \geq \ldots \geq \alpha_n$$
$$\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n}$$

In PrSS we can make a new notation of new height $k+1$ from sequences of height $\leq k$ as such:

Given a non-increasing *sequence of PrSS sequences* of height $\leq k$, e.g.

$$(0,1) \underset{\text{lex}}{\succeq} (0,0,0) \underset{\text{lex}}{\succeq} (0,0) \underset{\text{lex}}{\succeq} (0,0)$$

, we add 1 to each term in the sequence and join them up with 0s:

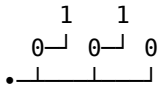
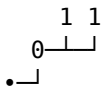
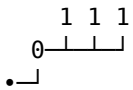
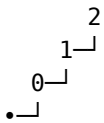
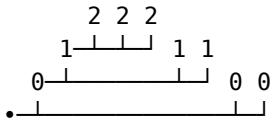
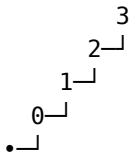
$$(0,1) \underset{\text{lex}}{\succeq} (0,0,0) \underset{\text{lex}}{\succeq} (0,0) \underset{\text{lex}}{\succeq} (0,0) \to (1,2) \underset{\text{lex}}{\succeq} (1,1,1) \underset{\text{lex}}{\succeq} (1,1) \underset{\text{lex}}{\succeq} (1,1)$$
$$(0,(1,2),0,(1,1,1),0,(1,1),0,(1,1))$$
$$(0,1,2,0,1,1,1,0,1,1,0,1,1)$$

Let's call this function f, which acts on *sequences of PrSS* and returns the concatenated sequence.

Since the lexicographic ordering of *sequences of PrSS* is also lexicographic (i.e. to compare 2 *sequences of PrSS*, we first compare the first sequence in both, then move onto the second if they are equal), for any two *sequences of PrSS* S_1 and S_2, $S_1 \underset{\text{lex}}{\prec} S_2$ if and only if $f(S_1) \underset{\text{lex}}{\prec} f(S_2)$.

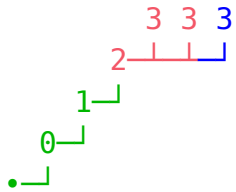
Here are some examples of the PrSS correspondence with ordinals. I added a third column for the PrSS "hydra" to show how elements in the sequence relate to nodes of the ordinal notation tree.

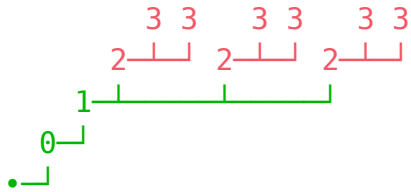
Ordinal	**Ordinal Notation**	**PrSS tree**	**PrSS**
0	0	•	$()$
1	ω^0	0 •└┘	(0)
2	$\omega^0 + \omega^0$	0 0 •└┴┘	$(0,0)$
3	$\omega^0 + \omega^0 + \omega^0$	0 0 0 •└┴┴┘	$(0,0,0)$
ω	ω^{ω^0}	1 0└┘ •└┘	$(0,1)$
$\omega+1$	$\omega^{\omega^0} + \omega^0$	1 0└┘ 0 •└┴──┘	$(0,1,0)$
$\omega+2$	$\omega^{\omega^0} + \omega^0 + \omega^0$	1 0└┘ 0 0 •└┴──┴┘	$(0,1,0,0)$
$\omega \cdot 2$	$\omega^{\omega^0} + \omega^{\omega^0}$	1 1 0└┘ 0└┘ •└┴──┘	$(0,1,0,1)$

Ordinal	Ordinal Notation	PrSS tree	PrSS
$\omega \cdot 2 + 1$	$\omega^{\omega^0} + \omega^{\omega^0} + \omega^0$		$(0, 1, 0, 1, 0)$
ω^2	$\omega^{\omega^0 + \omega^0}$		$(0, 1, 1)$
ω^3	$\omega^{\omega^0 + \omega^0 + \omega^0}$		$(0, 1, 1, 1)$
ω^ω	$\omega^{\omega^{\omega^0}}$		$(0, 1, 2)$
$\omega^{\omega^3+2} + 2$	$\omega^{\omega^{\omega^0 3} + \omega^0 2} + \omega^0 2$		$(0, 1, 2, 2, 2, 1, 1, 0, 0)$
ω^{ω^ω}	$\omega^{\omega^{\omega^{\omega^0}}}$		$(0, 1, 2, 3)$

We can see how this "hydra" behaves quite similarly to the Kirby-Paris Hydra we saw earlier:

$(0, 1, 2, 3, 3, 3)[3] = (0, 1, 2, 3, 3, 2, 3, 3, 2, 3, 3)$





3.2 Growth rate of $s[n]$

Since we have assigned a mapping from each PrSS sequence to an ordinal, let's compare PrSS expansion rules with the Hardy Hierarchy, using the Wainer Hierarchy for fundamental sequences. We make just one modification to H, that for a limit ordinal α, $H_\alpha(n) = H_{\alpha[n]}(n+1)$ instead of the usual $H_{\alpha[n]}(n)$. We can rewrite this as $H_\alpha(n) = H_{\alpha[n]+1}(n)$ since $H_{\alpha+1}(n) = H_\alpha(n+1)$.

PrSS	HH
$()[n] = n$	$H_0(n) = n$
$(0)[n] = ()[n+1]$	$H_1(n) = H_0(n+1)$
$(0, 0)[n] = (0)[n+1]$	$H_2(n) = H_1(n+1)$

PrSS	**HH**
$(0,1)[n] = \underbrace{(0,0,0,...,0)}_{n \text{ copies of } 0}[n+1]$	$H_\omega(n) = H_n(n+1)$
$(0,1,0)[n] = (0,1)[n+1]$	$H_{\omega+1} = H_\omega(n+1)$
$(0,1,0,1)[n] = \left(0,1,\underbrace{0,0,0,...,0}_{n \text{ copies of } 0}\right)[n+1]$	$H_{\omega\cdot 2}(n) = H_{\omega+n}(n+1)$
$(0,1,1)[n] = \underbrace{(0,1,...,0,1)}_{n \text{ copies of } (0,1)}[n+1]$	$H_{\omega^2}(n) = H_{\omega n}(n+1)$
$(0,1,2)[n] = \left(0,1,\underbrace{1,1,1,...,1}_{n \text{ copies of } 1}\right)[n+1]$	$H_{\omega^\omega}(n) = H_{\omega^n}(n+1)$

We see the correspondence between each PrSS sequence and its associated ordinal in our modified Hardy Hierarchy. Therefore the maximum growth rate of PrSS is $(0,1,2,3,...)[n]$ which corresponds to $H_{\varepsilon_0}(n)$.

07 Veblen Hierarchy

1 What lies beyond ε_0?

We previously spent a lot of time on ordinals $< \varepsilon_0$. We finally see what lies beyond ε_0. We first start with $\varepsilon_0 + 1$, then $\varepsilon_0 + 2, \varepsilon_0 + \omega, \varepsilon_0 + \omega^2, \varepsilon_0 + \omega^\omega$, until we get $\varepsilon_0 + \varepsilon_0 = \varepsilon_0 \cdot 2$. We then have the following ordinals, and their expression in Cantor Normal Form (CNF):

(Note that CNF applies to **all** ordinals, not just ordinals $< \varepsilon_0$. It is the *Ordinal Notations* discussed last chapter that have this restriction.)

$$\begin{aligned}
\varepsilon_0 \cdot 3 &= \omega^{\varepsilon_0} + \omega^{\varepsilon_0} + \omega^{\varepsilon_0} \\
\varepsilon_0 \cdot \omega &= \omega^{\varepsilon_0 + 1} \\
\varepsilon_0^2 &= \omega^{\varepsilon_0 \cdot 2} \\
\varepsilon_0^\omega &= \omega^{\omega^{\varepsilon_0 + 1}} \\
\varepsilon_0^{\varepsilon_0} &= \omega^{\omega^{\varepsilon_0 \cdot 2}} \\
\varepsilon_0^{\varepsilon_0^\omega} &= \omega^{\omega^{\omega^{\varepsilon_0 + 1}}} \\
\varepsilon_0^{\varepsilon_0^{\varepsilon_0}} &= \omega^{\omega^{\omega^{\varepsilon_0 \cdot 2}}} \\
\varepsilon_0^{\varepsilon_0^{\varepsilon_0^\omega}} &= \omega^{\omega^{\omega^{\omega^{\varepsilon_0 + 1}}}}
\end{aligned}$$

Note that while $\omega^{\varepsilon_0} = \varepsilon_0, \varepsilon_0^\omega \neq \varepsilon_0$.

$$\begin{aligned}
\omega^{\varepsilon_0} &= \bigcup_{\gamma < \varepsilon_0} (\omega^\gamma) = \overset{+}{\sup}\{\omega, \omega^\omega, \omega^\omega, \omega^{\omega^\omega}, \ldots\} = \varepsilon_0 \\
\varepsilon_0^\omega &= \bigcup_{\gamma < \omega} (\varepsilon_0^\gamma) > \varepsilon_0
\end{aligned}$$

The limit of stacking $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0 + 1}}}}$ is ε_1, or more formally:

$$\varepsilon_1 := \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0 + 1}, \omega^{\omega^{\varepsilon_0 + 1}}, \ldots\right\}$$

ε_1 can also be defined as:

$$\varepsilon_1 = \overset{+}{\sup}\left\{\varepsilon_0, \varepsilon_0^{\varepsilon_0}, \varepsilon_0^{\varepsilon_0^{\varepsilon_0}}, \ldots\right\}$$

However, you may be noticing that these are not the same sequence: $\varepsilon_0^{\varepsilon_0} = \omega^{\omega^{\varepsilon_0 \cdot 2}}$, which is not of the form $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0 + 1}}}}$. However, we can show that these two definitions lead to the same supremum of ε_1.

Proof

Let's fix $\varepsilon_1 = \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0 + 1}, \omega^{\omega^{\varepsilon_0 + 1}}, \ldots\right\}$. Suppose for contradiction that

$$\overset{+}{\sup}\left\{\varepsilon_0, \varepsilon_0^{\varepsilon_0}, \varepsilon_0^{\varepsilon_0^{\varepsilon_0}}, \ldots\right\} > \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \ldots\right\} = \varepsilon_1$$

That means there must exist a term $\varepsilon_0^{\cdot^{\cdot^{\varepsilon_0}}}$ greater than all $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0+1}}}}$. As seen earlier, we can express $\varepsilon_0^{\cdot^{\cdot^{\varepsilon_0}}}$ as some $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0 \cdot 2}}}}$. However, since $\varepsilon_0 + 1 \leq \varepsilon_0 \cdot 2 \leq \omega^{\varepsilon_0+1}$,

$$\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0+1}}}} \leq \omega^{\cdot^{\cdot^{\omega^{\varepsilon_0 \cdot 2}}}} \leq \omega^{\cdot^{\cdot^{\omega^{\omega^{\varepsilon_0+1}}}}}$$

So there cannot exist a term $\varepsilon_0^{\cdot^{\cdot^{\varepsilon_0}}}$ greater than all $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0+1}}}}$, and as such:

$$\overset{+}{\sup}\left\{\varepsilon_0, \varepsilon_0^{\varepsilon_0}, \varepsilon_0^{\varepsilon_0^{\varepsilon_0}}, \ldots\right\} = \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \ldots\right\} = \varepsilon_1$$

In fact, we can see that for any ordinal $\varepsilon_0 < \alpha < \varepsilon_1$:

$$\varepsilon_1 = \overset{+}{\sup}\left\{\omega^\alpha, \omega^{\omega^\alpha}, \omega^{\omega^{\omega^\alpha}}, \ldots\right\}$$

The fact that there are a wide variety of sequences with the same supremum means that beyond this point, fundamental sequences for these ordinals do not have an obvious "standard".

ε_1 is the second fixed point of $\omega^\alpha = \alpha$. Note that it is also a fixed point of $\varepsilon_0^\alpha = \alpha$.

We can repeat this process to get $\varepsilon_2 = \overset{+}{\sup}\left\{\varepsilon_1 + 1, \omega^{\varepsilon+1}, \omega^{\omega^{\varepsilon_1+1}}, \ldots\right\}$ and so on. ε_n is the $(1+n)$th fixed point of $\omega^\alpha = \alpha$, and ε_ω is the ωth fixed point of $\omega^\alpha = \alpha$. Formally:

$$\varepsilon_\omega = \overset{+}{\sup}\{\varepsilon_0, \varepsilon_1, \varepsilon_2, \ldots\} = \overset{+}{\sup}\{\varepsilon_n \mid n \in \mathbb{N}\}$$

and in general for a limit ordinal λ, $\varepsilon_\lambda = \overset{+}{\sup}\{\varepsilon_\alpha \mid \alpha < \lambda\}$

We can keep repeating until we get something like $\varepsilon_{\varepsilon_0}$ and $\varepsilon_{\varepsilon_{\varepsilon_0}}$. Eventually we reach:

$$\zeta_0 = \varepsilon_{\varepsilon_{\varepsilon_{\cdot^{\cdot^{\cdot}}}}} = \overset{+}{\sup}\left\{\varepsilon_0, \varepsilon_{\varepsilon_0}, \varepsilon_{\varepsilon_{\varepsilon_0}}, \ldots\right\}$$

ζ_0 is the first fixed point of $\varepsilon_\alpha = \alpha$, since $\varepsilon_{\zeta_0} = \varepsilon_{\left(\varepsilon_{\varepsilon_{\cdot^{\cdot^{\cdot}}}}\right)} = \varepsilon_{\varepsilon_{\varepsilon_{\cdot^{\cdot^{\cdot}}}}} = \zeta_0$. Note that it is also a fixed point of $\omega^\alpha = \alpha$, as $\omega^{\zeta_0} = \omega^{\varepsilon_{\zeta_0}} = \varepsilon_{\zeta_0} = \zeta_0$.

1.1 What lies beyond ζ_0?

We have the usual $\zeta_0 + 1$, all the way up to $\zeta_0 + \zeta_0 = \zeta_0 \cdot 2$. Then we have $\zeta_0 \cdot n$, all the way up to $\zeta_0 \cdot \omega = \omega^{\zeta_0+1}$. We can similarly construct:

$$\varepsilon_{\zeta_0+1} = \overset{+}{\sup}\left\{\zeta_0 + 1, \omega^{\zeta_0+1}, \omega^{\omega^{\zeta_0+1}}, \ldots\right\}$$

In order to climb to the next ζ, we need to nested deeper and deeper εs:

$$\varepsilon_{\varepsilon_{\zeta_0+1}} = \overset{+}{\sup}\left\{\varepsilon_{\zeta_0+1}, \omega^{\varepsilon_{\zeta_0+1}}, \omega^{\omega^{\varepsilon_{\zeta_0+1}}}, \ldots\right\}$$

$$\zeta_1 = \overset{+}{\sup}\left\{\zeta_0 + 1, \varepsilon_{\zeta_0+1}, \varepsilon_{\varepsilon_{\zeta_0+1}}, \varepsilon_{\varepsilon_{\varepsilon_{\zeta_0+1}}}, \ldots\right\}$$

When we finally reach $\zeta_{\zeta_{\zeta_{\cdot^{\cdot^{\cdot}}}}}$ we can define a new ordinal η_0 the first fixed point of $\zeta_\alpha = \alpha$. We can keep going, defining an even larger ordinal such that $\eta_\alpha = \alpha$ but eventually we will run our of symbols in this process. but at this point it's clear we need another way to express iteratively using fixed points.

1.2 Is there always a fixed point?

We have seen that $\omega^\alpha = \alpha$ has a fixed point, and we can convince ourselves it exists by informally saying $\omega^{\varepsilon_0} = \omega^{\omega^{\omega^{\cdot^{\cdot^{\cdot}}}}} = \omega^{\omega^{\omega^{\cdot^{\cdot^{\cdot}}}}} = \varepsilon_0$. We can similarly convince ourselves that the fixed point $\varepsilon_\alpha = \alpha$ also exists by saying $\varepsilon_{\zeta_0} = \varepsilon_{(\varepsilon_{\varepsilon_{\cdot^{\cdot^{\cdot}}}})} = \varepsilon_{\varepsilon_{\varepsilon_{\cdot^{\cdot^{\cdot}}}}} = \zeta_0$. But does a fixed point exist for every function? The answer is yes, if the function meets certain criteria:

Theorem 4 **(Veblen's Fixed Point Theorem)**

Definition 40

A **normal function** is a function $f : \mathrm{Ord} \mapsto \mathrm{Ord}$ such that:
1. f is **strictly increasing**: $f(\alpha) < f(\beta)$ whenever $\alpha < \beta$
2. f is **continuous**: if β is a limit ordinal, then $f(\beta) = \overset{+}{\sup}\{f(\alpha) \mid \alpha < \beta\}$

If a function is normal, then it has a fixed point $f(\beta) = \beta$. In fact there are infinitely many fixed points $f(\beta) = \beta$ that they make a proper class.

Proof

Let α be an arbitrary ordinal, and left f be a normal function. Now let's define β as:

$$\begin{aligned}\beta &= \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\} \\ &= \overset{+}{\sup}\{f^0(\alpha) = \alpha, f(\alpha), f(f(\alpha)), f^3(\alpha), \ldots\}\end{aligned}$$

Where $f^n(\alpha)$ denotes function iteration. We can now show that β is a fixed point:

$$\begin{aligned}f(\beta) &= f\left(\overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\}\right) \\ &= \overset{+}{\sup}\{f^{n+1}(\alpha) \mid n \in \mathbb{N}\} \\ &= \beta\end{aligned}$$

However, this assumes $\alpha \leq f(\alpha)$, which must hold, as otherwise if $\alpha > f(\alpha)$, we can make an infinite descending sequence:

$$\alpha > f(\alpha) > f(f(\alpha)) > f^3(\alpha) > \ldots$$

However, the ordinals are well-ordered, so such a sequence can't exist.

Since for any ordinal α, $\alpha \leq f(\alpha)$, we can conclude that $\alpha \leq \beta$. We can in fact show that $\beta = \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\}$ is the **least** fixed point f with the property $\alpha \leq \beta$:

Proof

Let γ be any fixed point of f, so $f(\gamma) = \gamma$, with the property $\alpha \leq \gamma$.

$$\begin{aligned}f(\alpha) &\leq f(\gamma) = \gamma \\ f(f(\alpha)) &\leq f(f(\gamma)) = \gamma \\ f^n(\alpha) &\leq \gamma \\ \beta = \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\} &\leq \gamma\end{aligned}$$

Therefore there is a proper class of fixed points of f, no matter what f is.

We can define a function to find these fixed points:

Definition 41

The **derivative** of normal functions f, denoted $f'(\alpha)$, is a function that enumerates over the fixed points of f.

$$f'(\alpha) = \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\} \leq \gamma$$

Example 26

The derivative of $f(\alpha) = \omega^\alpha$ is $f'(\alpha) = \varepsilon_\alpha$, e.g. ε_0 is the 1st fixed point of $f(\alpha) = \omega^\alpha$

We can generalize this to define a fixed point of multiple functions:

Definition 42

For a family of the function $f_1, f_2, ..., f_k$, the *joint fixed point* of *all* $f_1, f_2, ..., f_k$ is defined as the fixed point of

$$f(\alpha) := f_1(f_2(...f_k(\alpha)))$$

So the $(1+\alpha)^{\text{th}}$ fixed point of $f_1, f_2, ..., f_k$ is:

$$f'(\alpha) = \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\}$$

2 2-argument Veblen function

Definition 43

The 2-argument/binary Veblen function $\boldsymbol{\varphi(\alpha, \beta)}$ is defined as such:
- $\varphi(0, \gamma) := \omega^\gamma$
- $\varphi(\beta, \gamma)$ for $\alpha \neq 0$ is the $(1+\gamma)^{\text{th}}$ fixed point of the functions $\varphi(\alpha, \bullet) = \bullet$ for $\alpha < \beta$

Since we will be dealing with functions with multiple arguments, it's hard to tell which argument is the one being fixed, so I denote the "dummy variable" of the fixed point to be $\bullet$.

Example 27

For an ordinal α:

$$\begin{aligned}\omega^\alpha &= \varphi(0, \alpha)\\ \varepsilon_\alpha &= \varphi(1, \alpha)\\ \zeta_\alpha &= \varphi(2, \alpha)\\ \eta_\alpha &= \varphi(3, \alpha)\end{aligned}$$

Show that $\varepsilon_0 = \varphi(1, 0)$, and $\zeta_2 = \varphi(2, 2)$

1. $\varphi(1, 0)$ is the $(1+0)^{\text{th}}$ fixed point of the function $\varphi(\gamma, \bullet) = \bullet$ for $\gamma < 1$ (i.e. $\gamma = 0$)

$\varphi(1,0)$ is the 1st fixed point of the function $\varphi(0,\bullet) = \omega^{\bullet} = \bullet$, and the first fixed point of $\omega^{\bullet} = \bullet$ is ε_0.

2. $\varphi(2,2)$ is the $(1+2)^{\text{th}}$ fixed point of the function $\varphi(\gamma,\bullet) = \bullet$ for $\gamma < 2$

   $\varphi(2,2)$ is the 3rd fixed point of the functions $\varphi(0,\bullet) = \omega^{\bullet} = \bullet$ and $\varphi(1,\bullet) = \varepsilon_{\bullet} = \bullet$, and ζ_2 is the third ordinal that is a fixed point for *both* functions. (ζ_0, ζ_1 are the first and second).

Let's go through some basic properties of the Veblen function:

Proposition 13

1. If $\alpha < \beta$, then $\varphi(\alpha, \varphi(\beta,\gamma)) = \varphi(\beta,\gamma)$
2. If $\alpha < \beta$, then $\varphi(\gamma,\alpha) < \varphi(\gamma,\beta)$

Both properties are just consequences of the definition provided, so convince yourself that they are true.

Example 28

Which is bigger? $\varphi(2,\varphi(1,0))$ or $\varphi(3,0)$?

Of course we can convert them into ordinals that we know (ζ_ω and η_0) but we don't have notation that keeps up.

$\varphi(3,0)$ can be written as $\varphi(1,\varphi(3,0))$, which can then be written as $\varphi(2,\varphi(1,\varphi(3,0)))$ (Property 1)

Then we have:

$$\begin{aligned} 0 &< \varphi(3,0) \\ \varphi(1,0) &< \varphi(1,\varphi(3,0)) \\ \varphi(2,\varphi(1,0)) &< \varphi(2,\varphi(1,\varphi(3,0))) = \varphi(3,0) \end{aligned}$$

Therefore $\varphi(3,0)$ is bigger.

In the previous section, we defined ε_1 as:

$$\begin{aligned} \varepsilon_1 &= \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \ldots\right\} \\ &= \overset{+}{\sup}\{\varphi(1,0)+1, \varphi(0,\varphi(1,0)+1), \varphi(0,\varphi(0,\varphi(1,0)+1)), \ldots\} \\ &= \overset{+}{\sup}\{\varphi^n(0,\varphi(1,0)+1) | n \in \mathbb{N}\} \\ &= \varphi(1,1) \end{aligned}$$

We also mentioned that instead of $\varphi(1,0)+1$ (i.e. $\varepsilon_0 + 1$), we can replace that with any ordinal α such that $\varphi(1,0) < \alpha < \varphi(1,1)$ (i.e. $\varepsilon_0 < \alpha < \varepsilon_1$). However, it is a bit self-referential to use $\varphi(1,1)$ in a condition that defines $\varphi(1,1)$. We can instead prove a similar statement for all $\varphi(\alpha+1,\beta+1)$:

Proposition 14

$\varphi(\alpha+1,\beta+1)$ is the first fixed point of the function $f(\bullet) = \varphi(\alpha,\varphi(\alpha+1,\beta)+\bullet) = \bullet$

Proof

Let ξ be the first (i.e. *least*) fixed point of f, i.e., $f(\xi) = \varphi(\alpha, \varphi(\alpha+1, \beta) + \xi) = \xi$. We now show that $\xi = \varphi(\alpha+1, \beta+1)$

Since $\varphi(\alpha, \gamma) \geq \gamma$ for all γ, we can express ξ as:

$$\xi = \varphi(\alpha, \varphi(\alpha+1, \beta) + \xi) \geq \varphi(\alpha+1, \beta) + \xi$$

This gives us $\xi \geq \varphi(\alpha+1, \beta) + \xi$. Since obviously $\xi \not> \varphi(\alpha+1, \beta) + \xi$, we have:

$$\xi = \varphi(\alpha+1, \beta) + \xi$$

, and for that to be true, $\xi > \varphi(\alpha+1, \beta)$. Applying $\varphi(\alpha, \bullet)$ on both sides we get

$$\begin{aligned} \varphi(\alpha, \xi) &= \varphi(\alpha, \varphi(\alpha+1, \beta) + \xi) \\ &= \xi \end{aligned}$$

Therefore ξ is a fixed point of $\varphi(\alpha, \bullet)$, i.e., $\xi = \varphi(\alpha+1, \gamma)$ for some ordinal γ. And since $\xi > \varphi(\alpha+1, \beta)$, minimally $\xi \geq \varphi(\alpha+1, \beta+1)$.

We also notice that $\varphi(\alpha+1, \beta+1)$ is a fixed point of f:

$$\begin{aligned} f(\varphi(\alpha+1, \beta+1)) &= \varphi(\alpha, \varphi(\alpha+1, \beta) + \varphi(\alpha+1, \beta+1)) \\ &= \varphi(\alpha, \varphi(\alpha+1, \beta+1)) \\ &= \varphi(\alpha+1, \beta+1) \end{aligned}$$

Since ξ is the *least* fixed point of f, $\xi \leq \varphi(\alpha+1, \beta)$. Since we have just shown earlier that $\xi \geq \varphi(\alpha+1, \beta+1)$, we get:

$$\xi = \varphi(\alpha+1, \beta+1)$$

Note that with this definition our ε_1 sequence would be translated as:

$$\begin{aligned} \varphi(1,1) &= \overset{+}{\sup}\{f^{n(1)} | n \in \mathbb{N}\} \\ &= \overset{+}{\sup}\{\varphi(0, \varphi(1,0)+1), \varphi(0, \varphi(1,0) + \varphi(0, \varphi(1,0)+1)), \ldots\} \\ &= \overset{+}{\sup}\left\{\omega^{\varepsilon_0+1}, \omega^{\varepsilon_0+\omega^{\varepsilon_0+1}}, \omega^{\varepsilon_0+\omega^{\varepsilon_0+\omega^{\varepsilon_0+1}}}, \ldots\right\} \\ &= \overset{+}{\sup}\left\{\omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \omega^{\omega^{\omega^{\varepsilon_0+1}}}, \ldots\right\} \end{aligned}$$

Now that we have defined the 2-argument Veblen function, what is the largest ordinal we can make? After η_0 we have $\varphi(4,0), \varphi(10,0)$, or even $\varphi(10000,0)$. What about $\varphi(\omega,0)$? We can think of it as the first fixed point for *all* $\varphi(n, \bullet) = \bullet$ for every $n \in \mathbb{N}$, or as the supremum of all $\varphi(n,0)$:

$$\varphi(\omega, 0) = \overset{+}{\sup}\{\varphi(n,0) \mid n \in \mathbb{N}\}$$

It should be obvious to deduce that it is also the supremum of all $\varphi(n,1)$ too, i.e., the value of the second argument does not matter. In general, for a limit ordinal λ,

$$\varphi(\lambda, 0) = \overset{+}{\sup}\{\varphi(\alpha, 0) | \alpha < \lambda\}$$

Continuing where we left off, we have $\varphi(\omega,0) = \varphi(\varphi(1,0),0)$, and we can keep nesting $\varphi(\varphi(\varphi(1,0),0),0)$, $\varphi(\varphi(\varphi(\varphi(1,0),0),0),0)$, until we reach the (first) fixed point of the Veblen

function: $\Gamma_0 = \varphi(\Gamma_0, 0)$. Γ_0 is known as the **Feferman–Schütte ordinal**, which can also be expressed as:

$$\Gamma_0 = \overset{+}{\sup}\{\varphi(1,0), \varphi(\varphi(1,0),0), \varphi(\varphi(\varphi(1,0),0),0), ...\}$$

This is the largest ordinal we can reach with the two-argument veblen function.

3 Veblen Normal Form and Fundamental Sequences

Much like how all ordinals $< \varepsilon_0$ can be expressed in Cantor Normal Form, all ordinals $< \Gamma_0$ can be expressed in:

Theorem 5 (**Veblen Normal Form**)

Every ordinal $\alpha < \Gamma_0$, can be uniquely written in the following form:

$$\varphi(\beta_1, \gamma_1) + \varphi(\beta_2, \gamma_2) + ... + \varphi(\beta_k, \gamma_k)$$

where $\varphi(\beta_1, \gamma_1) \geq \varphi(\beta_2, \gamma_2) \geq ... \geq \varphi(\beta_k, \gamma_k)$ and $\gamma_i < \varphi(\beta_i, \gamma_i)$ for $i \in \{1, ..., k\}$

Definition 44

The **Veblen Hierarchy** is a system of fundamental sequences for **limit** ordinals $\alpha < \Gamma_0$ when expressed in Veblen Normal Form:

1. $(\varphi(\beta_1, \gamma_1) + \varphi(\beta_2, \gamma_2) + ... + \varphi(\beta_k, \gamma_k))[n] := \varphi(\beta_1, \gamma_1) + \varphi(\beta_2, \gamma_2) + ... + \varphi(\beta_k, \gamma_k)[n]$
2. $\varphi(0, \gamma + 1) := \omega^\gamma \cdot n$
3. $\varphi(\beta + 1, 0)[n] := \varphi^n(\beta, 0)$
4. $\varphi(\beta + 1, \gamma + 1)[n] := \varphi^n(\beta, \varphi(\beta + 1, \gamma) + 1)$
5. $\varphi(\beta, \gamma)[n] := \varphi(\beta, \gamma[n])$ when γ is a limit ordinal
6. $\varphi(\beta, 0)[n] := \varphi(\beta[n], 0)$ when β is a limit ordinal
7. $\varphi(\beta, \gamma + 1)[n] := \varphi(\beta[n], \varphi(\beta, \gamma) + 1)$ when β is a limit ordinal

In rule 3 and 4, $\varphi^n()$ refers to function iteration, so $\varphi^n(\beta, \gamma) = \varphi(\beta, \varphi^{n-1}(\beta, \gamma))$ and $\varphi^1(\beta, \gamma) = \varphi(\beta, \gamma)$

Example 29

- $\varphi(2, 0)[n]$

$$\begin{aligned}
\varphi(2,0)[1] &= \varphi^1(1,0) = \varphi(1,0) = \varepsilon_0 \\
\varphi(2,0)[2] &= \varphi^2(1,0) = \varphi(1, \varphi(1,0)) = \varepsilon_{\varepsilon_0} \\
\varphi(2,0)[3] &= \varphi^3(1,0) = \varphi(1, \varphi(1, \varphi(1,0))) = \varepsilon_{\varepsilon_{\varepsilon_0}}
\end{aligned}$$

- $\varphi(2, 1)[n]$

$$\begin{aligned}
\varphi(2,1)[1] &= \varphi^1(1, \varphi(2,0) + 1) = \varepsilon_{\zeta_0 + 1} \\
\varphi(2,1)[2] &= \varphi^2(1, \varphi(2,0) + 1) = \varphi(1, \varphi(1, \varphi(2,0) + 1)) = \varepsilon_{\varepsilon_{\zeta_0 + 1}} \\
\varphi(2,1)[3] &= \varphi^3(1, \varphi(2,0) + 1) = \varphi(1, \varphi(1, \varphi(1, \varphi(2,0) + 1))) = \varepsilon_{\varepsilon_{\varepsilon_{\zeta_0 + 1}}}
\end{aligned}$$

- $\varphi(\omega, 0)[n]$

$$\varphi(\omega, 0)[n] = \varphi(\omega[n], 0) = \varphi(n, 0)$$

- $\varphi(\omega, 1)[n]$

$$\varphi(\omega, 1)[n] = \varphi(\omega[n], \varphi(\omega, 0) + 1) = \varphi(n, \varphi(\omega, 0) + 1)$$

4 Multivariate Veblen function

4.1 3-argument veblen function

So how do we go beyond Γ_0? Γ_0 is the 1st fixed point of $\varphi(\bullet, 0) = \bullet$. Γ_1 is therefore the 2nd fixed point, and Γ_2 is the 3rd, and so on. This "Gamma function" can be thought of as $\Gamma_\beta = (1+\beta)^{\text{th}}$ fixed point of $\varphi(\bullet, 0) = \bullet$. So we can make Γ_ω, Γ_{ε_0}, or even Γ_{Γ_0}. We have seen this pattern before – eventually we reach the *Gamma fixed point* of $\Gamma_\bullet = \bullet$, which can be though of as an infinite nesting of Gammas: $\Gamma_{\Gamma_{\Gamma_{\cdot^{\cdot^{\cdot}}}}}$.

We can then iterate over the fixed points of $\Gamma_\bullet = \bullet$, but we can systematize this with the *3-argument veblen function*. We let $\Gamma_0 = \varphi(1, 0, 0)$, and then $\Gamma_1 = \varphi(1, 0, 1)$. In general $\Gamma_\beta = \varphi(1, 0, \beta)$, and first Gamma fixed point $\Gamma_{\Gamma_{\Gamma_{\cdot^{\cdot^{\cdot}}}}} = \varphi(1, 1, 0)$ can be denoted as the first fixed point of $\varphi(1, 0, \bullet) = \bullet$.

We can create a definition of the 3-argument veblen function as such:

Definition 45

- $\varphi(0, \beta, \gamma) = \varphi(\beta, \gamma)$, converting from the 3-argument veblen to the 2-argument veblen
- $\varphi(\alpha, 0, \gamma)$ is the $(1+\gamma)^{\text{th}}$ fixed point of $\varphi(\alpha', \bullet, 0) = \bullet$ for all $\alpha' < \alpha$
- $\varphi(\alpha, \beta, \gamma), \beta \neq 0$ is the $(1+\gamma)^{\text{th}}$ fixed point of $\varphi(\alpha, \beta', \bullet)$ for all $\beta' < \beta$

Example 30

$\varphi(2, 0, 0)$ is the 1st fixed point of $\varphi(\alpha', \bullet, 0) = \bullet$ for all $\alpha' < \alpha = 2$, so $\alpha' = \{0, 1\}$
- $\varphi(0, \bullet, 0) = \bullet$
- $\varphi(1, \bullet, 0) = \bullet$

4.2 Extension to finitely many arguments

Now what about a 4-argument veblen function? or an n-argument veblen function? We can extend it to any number of arguments as such:

Definition 46

Let S be an arbitrary string of ordinals, and O be a string of 0s.
- $\varphi(\gamma) = \gamma$
- $\varphi(O, S) = \varphi(S)$, i.e., you can remove zeroes from the left
- $\varphi(S, \beta, O, \gamma)$ is the $(1+\gamma)^{\text{th}}$ fixed point of $\varphi(S, \alpha, \bullet, O) = \bullet$ for all $\alpha < \beta$

Example 31

1. $\varphi(1, 0, 0, 0)$

We have $S = \text{empty}, \beta = 1, O = (0,0), \gamma = 1$:

$$\varphi\left(\underbrace{1}_{\beta}, \underbrace{0,0}_{O}, \underbrace{0}_{\gamma}\right)$$

As such $\varphi(1,0,0,0)$ is the $(1+0)^{\text{th}}$ fixed point of $\varphi(\alpha, \bullet, 0, 0) = \bullet$ for all $\alpha < 1$, i.e. $\alpha = 0$. Rephrasing, we have:

$\varphi(1,0,0,0)$ is the 1st fixed point of $\varphi(\bullet, 0, 0) = \bullet$.

2. $\varphi(1,2,3,4)$ We have $S = (1,2), \beta = 3, O = \text{empty}, \gamma = 4$

   As such $\varphi(1,2,3,4)$ is the $(1+4)^{\text{th}}$ fixed point of $\varphi(1,2,\alpha,\bullet) = \bullet$ for all $\alpha < 3$. Rephrasing, we have:

   $\varphi(1,2,3,4)$ is the 5th fixed point of $\varphi(1,2,\alpha,\bullet) = \bullet$ for $\alpha \in \{0,1,2\}$.

With this, we can define ordinals like $\varphi(1,0,0,0)$, $\varphi(1,0,0,0,0)$, and so on. Taking this to its limit yields the **Small Veblen Ordinal**:

$$\text{SVO} = \overset{+}{\sup}\{\varphi(1), \varphi(1,0), \varphi(1,0,0), \varphi(1,0,0,0), \varphi(1,0,0,0,0), \ldots\}$$

We also have an "Extended Veblen Normal Form":

Theorem 6

Every ordinal $\alpha < \text{SVO}$, can be uniquely written in the following form:

$$\varphi(s_1) + \varphi(s_2) + \ldots + \varphi(s_k)$$

where:
- $\varphi(s_1) \geq \varphi(s_2) \geq \ldots \geq \varphi(s_k)$
- s_i is an arbitrary string of ordinals $\alpha_{i,1}, \alpha_{i,2}, \ldots, \alpha_{i,k_i}$ where $i \in \{1, \ldots, k\}$
- $\alpha_{i,1} > 0$ and $\alpha_{i,j} < \varphi(s_i)$ for all $i \in \{1, \ldots, k\}$ and $j \in \{1, \ldots, k_i\}$

However, I'm going to skip straight to the veblen function with *transfinitely* many arguments, as that generalizes to finitely many arguments too.

5 Transfinitary Veblen function

5.1 Finitely Supported functions

Definition 47

A function $f : X \mapsto Y$ if a **finitely supported** if there are only *finitely many* values of $x \in X$ such that $f(x) \neq 0$ (assuming $0 \in Y$).

We call the set of values $x \in X$ such that $f(x) \neq 0$ the *support* of the function. $\{x | x \in X \text{ and } f(x) \neq 0\})$

If $f : X \mapsto Y$ is finitely supported, we write it as $f : X \mapsto_0 Y$.

Example 32

$$f(\beta) = \begin{cases} 2 \text{ if } \beta = 0 \\ 3 \text{ if } \beta = 1 \\ 0 \text{ else} \end{cases}$$

is an example of a finitely supported function from Ord $\mapsto_0$ Ord

We use finitely supported functions for this as our veblen function will now *take in* a finitely supported function f : Ord $\mapsto_0$ Ord, so our veblen function can be thought of as φ : (Ord $\mapsto_0$ Ord) $\mapsto$ Ord.

Example 33

In our previous example, we defined an f where $f(0) = 2, f(1) = 3, f(\text{anything else}) = 0$. Then

$$\varphi(f) = \varphi(f(1), f(0)) = \varphi(3, 2)$$

and in general, the arguments are arranged "right to left":

$$\varphi(f) = \varphi(..., f(2), f(1), f(0))$$

The easiest way to create a new finitely supported function from another is to "update" one of their values:

Definition 48

For a finitely supported function f, $f[\alpha@\beta]$ is defined as such:

$$f[\alpha@\beta](\xi) := \begin{cases} f(\xi) & \text{if } \xi \neq \beta \\ \alpha & \text{if } \xi = \beta \end{cases}$$

You can think of $f[\alpha@\beta]$ as "f with α at β".

As shorthands:
1. If $f(\xi) = 0$ for all ξ, i.e., f is the "zero function", then $f[\alpha@\beta]$ can be written as just $[\alpha@\beta]$
2. $f[\alpha_1@\beta_1][\alpha_2@\beta_2]...[\alpha_n@\beta_n]$ can be written as $f[\alpha_1@\beta_1, \alpha_2@\beta_2, ..., \alpha_n@\beta_n]$

Proposition 15

The set of all finitely supported functions Ord $\mapsto_0$ Ord is well-ordered as such:

For two finitely supported functions f, g : Ord $\mapsto_0$ Ord, $f < g$ if for some ordinal α:
1. $f(\beta) = g(\beta)$ for all $\beta < \alpha$, and
2. $f(\alpha) < g(\alpha)$

This should be fairly easy to show, it is quite similar to how $\mathbb{N}^2$ is well-ordered.

Definition 49

The **degree** $\deg(f)$ of a finitely supported function $f : X \mapsto_0 Y$ is the smallest $d \in X$ such that $f(d) \neq 0$ (i.e. the smallest element of X that is in the support of f)

5.2 Definition

Definition 50

The multivariate Veblen function is the function $\varphi : (\mathrm{Ord} \mapsto_0 \mathrm{Ord}) \mapsto \mathrm{Ord}$ defined as follows:

Let $f : \mathrm{Ord} \mapsto_0 \mathrm{Ord}$, and let $g = f[0@0]$. If g is the zero function, we set $\varphi(f) = \omega^{f(0)}$.

Otherwise, let $d = \deg(g)$. We define $\varphi(f)$ as the $(1 + f(0))^{\text{th}}$ common fixed point of the functions $\varphi(g[\alpha@d, \bullet@d']) = \bullet$ for all $\alpha < f(d)$ and $d' < d$.

Example 34

$\varphi(1, 1, 0) = \varphi(f)$ where f is:

$$f(\xi) = \begin{cases} 0 \text{ if } \xi = 0 \\ 1 \text{ if } \xi = 1 \\ 1 \text{ if } \xi = 2 \\ 0 \text{ else} \end{cases}$$

$g = f[0@0]$ basically sets $g(0) = 0$, which doesnt change anything since $f(0)$ is already 0. So g is not the zero function.

$d = \deg(f) = 1$. So $\varphi(f)$ is the $(1 + f(0))^{\text{th}} = 1^{\text{st}}$ common fixed point of: $\varphi(g[\alpha@d, \bullet@d']) = \bullet$ for all $\alpha < f(d) = 1$ and $d' < d = 1$. So we are basically stuck with $\alpha = 0$ and $d' = 0$, so $\varphi(1, 1, 0)$ is the first fixed point of $\varphi(f[0@1, \bullet@0]) = \varphi(1, 0, \bullet) = \bullet$

Example 35

$\varphi([1@\omega]) = \varphi(f)$ where f is:

$$f(\xi) = \begin{cases} 1 \text{ if } \xi = \omega \\ 0 \text{ else} \end{cases}$$

$g = f[0@0] = f$ which doesn't change anything since $f(0) = 0$. $\deg(f) = \omega$, so from the definition, $\varphi(f)$ is the 1st common fixed point of all functions $\varphi(g[\alpha@d, \bullet@d']) = \bullet$ for all $\alpha < f(d) = 1$ and $d' < d = \omega$. Plugging this in yields $\varphi(g[0@\omega, \bullet@d']) = \bullet$ for all $d' < \omega$, or put another way it is the common fixed points of all $\varphi(1@n)$ for $n \in \mathbb{N}$.

This is equivalent to the *Small Veblen Ordinal*.

But how do we even know that these fixed points exist? We need to first show that the multivariate veblen are all normal functions.

Proposition 16

If $f : \mathrm{Ord} \mapsto_0 \mathrm{Ord}$, and $\deg(f) \geq d$, then $g(\alpha) = \varphi(f[\alpha@d])$ is a non-negative normal function.

Proof

[WIP]

Now that we have extended the veblen function yet again, what is the new limit? We can construct $\mathrm{SVO} = \varphi([1@\omega])$, but why stop at ω? We can have $\varphi([1@\Gamma_0])$, $\varphi([1@\mathrm{SVO}]) = \varphi([1@\varphi([1@\omega])])$, or even $\varphi([1@\varphi([1@\varphi([1@\omega])])])$. I think we have seen this pattern enough times by now – we can

make a fixed point $\varphi([1@\bullet]) = \bullet$, and the (first) ordinal that satisfies is known as the **Large Veblen Ordinal** LVO. If we let $f(\alpha) = \varphi([1@\alpha])$, it can also be written as:

$$\begin{aligned}\text{LVO} &= \overset{+}{\sup}\{f^n(0) \mid n \in \mathbb{N}\} \\ &= \overset{+}{\sup}\{f(0), f(f(0)), f(f(f(0))), ...\} \\ &= \overset{+}{\sup}\{\varphi([1@0]) = \omega, \varphi([1@\varphi[1@0]]) = \text{SVO}, \varphi([1@\varphi[1@\varphi([1@0])]]), ...\}\end{aligned}$$

We do have to first show that $\varphi([1@\alpha])$ is normal before we know that such a fixed point exists:

Proposition 17

The function $f(\alpha) = \varphi([1@\alpha])$ is normal.

Proof

We need to show that it is:

1. strictly increasing

   $\varphi([1@y]) = \varphi([\varphi([1@y])@x]) > \varphi([1@x])$ (why? [WIP])
2. continuous [WIP]

Theorem 7 (**Tranfinitary Veblen Normal Form**)

All ordinals $\alpha < \text{LVO}$ can be expressed as

Definition 51

The fundamental sequences for the multivariate veblen function is: [WIP]

5.3 Schütte's Klammersymbolen

This is another way to write the veblen function with transfinitely many arguments. If we have a finitely covered function f like so:

$$f(\xi) = \begin{cases} \alpha_0 & \text{if } \xi = 0 \\ \alpha_1 & \text{if } \xi = 1 \\ \quad\vdots & \\ \alpha_\beta & \text{if } \xi = \beta \\ 0 & \text{else} \end{cases}$$

We can "decompose" it to $\left[\alpha_0@0, \alpha_1@1, ..., \alpha_\beta@\beta\right]$. In Schütte's Klammersymbolen, we write this as:

$$\begin{pmatrix} \alpha_\beta & ... & \alpha_1 & \alpha_0 \\ \beta & ... & 1 & 0 \end{pmatrix}$$

So for example, the Small Veblen Ordinal would be $\varphi\binom{1}{\omega}$, and the Large Veblen Ordinal would be represented as the following fixed point:

$$\mathrm{LVO} = \varphi\begin{pmatrix} 1 \\ \mathrm{LVO} \end{pmatrix}$$

6 Dimensional Veblen

Some fellas made another extension to the Veblen function. I won't cover it here but you can check it out here: https://arxiv.org/pdf/2310.12832

08 Buchholz's Ordinal Collapsing Function

Buchholz's **Ordinal Collapsing Function** (BOCF) is a way for us to create even larger (countable) ordinals. Ironically, to make these larger ordinals, we will be using even larger *uncountable* ordinals to get there.

But first, how exactly are ordinals of higher cardinality ($\omega_1, \omega_2, \ldots$) defined? Let's first define the *successor cardinal* of $\aleph_0$, $\aleph_1$, and its assigned ordinal ω_1 as **the set of all ordinals whose cardinality $\leq \aleph_0$**:

$$\aleph_1 = \omega_1 = \{\alpha \in \mathrm{Ord} : |\alpha| \leq \aleph_0\}$$

where $|\alpha|$ represents the cardinality of an ordinal α. Similarly, we have the cardinal $\aleph_2$ and its assigned ordinal ω_2 to be the set of all ordinals whose cardinality is $\leq \aleph_1$. We can repeat this to make $\aleph_3, \aleph_4, \ldots$. For a limit ordinal λ, we define $\aleph_\lambda$ as such:

$$\aleph_\lambda = \bigcup_{\beta<\lambda} \aleph_\beta$$

Definition 52 (**Buchholz's ψ function**)

Let Ω_ν be defined as such:

$$\Omega_\nu := \begin{cases} 1 & \text{if } \nu = 0 \\ \omega_\nu & \text{if } \nu > 0 \end{cases}$$

Buchholz's ordinal collapsing function ψ and the set $C_\nu(\alpha)$ is defined as such:

1. $\Omega_\nu \subseteq C_\nu(\alpha)$
2. For any two ordinals $\xi, \eta \in C_\nu(\alpha)$, their sum $\xi + \eta \in C_\nu(\alpha)$
3. For any ordinal $\xi \in C_\nu(\alpha)$, as long as $\xi < \alpha$, then $\psi_\mu(\xi) \in C_\nu(\alpha)$ for all $\mu \leq \omega$
4. $\psi_\nu(\alpha) = \min\{\gamma \in \mathrm{Ord} \mid \gamma \notin C_\nu(\alpha)\}$, i.e., the smallest ordinal not inside $C_\nu(\alpha)$

This is a lot to unpack, so let's first start with $\nu = 0$:

Example 36

Let's first find what $\psi_0(0)$ is.

In order to figure out what $\psi_0(0)$, we first need to know what's inside $C_0(0)$. Going down the list, we have:

1. $\Omega_0 = 1 = \{0\} \subseteq C_0(0)$

   so currently $C_0(0) = \{0\}$

2. For any two ordinals in $C_0(0)$, their sum is also in $C_0(0)$

   Another way to interpret this statement is to say that you can take a collection of ordinals in $C_0(0)$ and apply addition a finite number of times on them. In this case, all we have is 0, so nothing new gets added here.

3. For any ordinal $\xi \in C_0(0)$, as long as $\xi < 0$, then $\psi_\mu(\xi) \in C_0(0)$ for all $\mu \leq \omega$

In this case, there are no ordinals $\xi < 0$ so, nothing new get added here either.

4. $\psi_0(0)$ is the smallest ordinal *not* contained inside $C_0(0) = \{0\}$, which is just 1.

In fact, we can generalize this and show that:

Lemma 2

$\psi_\nu(0) = \Omega_\nu$

Proof

From the definition, $\Omega_\nu \subseteq C_\nu(0)$, i.e. for all $\alpha \in \Omega_\nu, \alpha \in C_\nu(0)$.

For all $\alpha, \beta \in \Omega_\nu, \alpha + \beta \in \Omega_\nu$, so adding doesn't generate us anything new.

Since there are no ordinals $\xi < 0$, we can't use ψ to generate us anything new.

So $C_\nu(0) = \{\gamma \in \text{Ord} \mid \gamma < \Omega_\nu\}$. Then $\psi_\nu(0)$ is the smallest ordinal not in $C_\nu(0)$ which is Ω_ν.

Example 37

Let's proceed to find the value of $\psi_0(1)$:

1. $\Omega_0 \subseteq C_0(0)$, so we once again have $0 \in C_0(0)$.
2. Adding doesn't do anything

The difference this time is that we can use $\psi_0(0)$:

3. For any ordinal $\xi \in C_0(0)$, as long as $\xi < 1$, then $\psi_{\mu(\xi)} \in C_0(0)$ for all $\mu \leq \omega$

   The only valid values of ξ is just 0, so we have $\psi_0(0), \psi_1(0), \ldots, \psi_{\omega(0)} \in C_0(0)$. $\psi_0(0) = 1 \in C_0(0)$, so we finally have something new. We haven't gotten to $\psi_1(0), \psi_2(0), \ldots, \psi_{\omega(0)}$ yet, and we'll ignore them for now.
2. We can now use addition to show that $1 + 1 = 2 \in C_0(0), 2 + 1 = 3 \in C_0(0)$, and so on to show that all natural numbers $\mathbb{N} \subseteq C_0(0)$. In fact, $\mathbb{N} = C_0(0)$.
4. $\psi_0(1)$ is the smallest ordinal not in $C_0(0)$, which is ω

1 Properties

Before we continue further, let's look at some properties about this function. Let's first introduce the concept of an **additively principal** ordinal:

Definition 53

An **additively principal** ordinal α is any non-zero ordinal such that for any $\beta, \gamma < \alpha$, $\beta + \gamma < \alpha$.

We denote the *class* of all additively principal numbers as P. P is precisely all ordinals of the form ω^ξ, where ξ is some ordinal.

Formally, we can define P as such:

$$\begin{aligned} P &:= \{\alpha \in \text{Ord} \mid 0 < \alpha \text{ and for all } \beta, \gamma < \alpha, \beta + \gamma < \alpha\} \\ &= \left\{\omega^\xi \mid \xi \in \text{Ord}\right\} \end{aligned}$$

We can also define a function $P(\alpha)$ that "decomposes" an ordinal α into its additively principal "components":

Definition 54

We define $P(\alpha)$ as:
1. $P(0) := \emptyset$
2. For $\alpha > 0$, there are uniquely determined $\alpha_0, \alpha_1, ..., \alpha_n \in P$ with $\alpha = \alpha_0 + \alpha_1 + ... + \alpha_n$.

   We set $P(\alpha) := \{\alpha_0, \alpha_1, ..., \alpha_n\}$

This notion is useful as the Buchholz function ψ, as each $C_\nu(\alpha)$ represents all ordinals that can be derived from addition and the ψ function itself:

Lemma 3

1. If $\zeta, \eta \in C_\nu(\alpha)$, then $\zeta + \eta \in C_\nu(\alpha)$
2. An ordinal $\gamma \in C_\nu(\alpha)$ if and only if $P(\gamma) \subseteq C_\nu(\alpha)$
3. If $\zeta + \eta \in C_\nu(\alpha)$, then $\eta \in C_\nu(\alpha)$

Proof

1. In the definition
2. Since $\gamma = \gamma_0 + \gamma_1 + ... + \gamma_n$ where $\gamma_i \in P(\gamma)$, and $P(\gamma) \subseteq C_\nu(\alpha)$, then $\gamma_i \in C_\nu(\alpha)$ for all $i \leq n$, and as such, we can "reconstruct" γ from elements of $C_\nu(\alpha)$:
$$\gamma = \gamma_0 + ... + \gamma_n \in C_\nu(\gamma)$$
3. Since $\xi + \eta \in C_\nu(\alpha)$, we have $P(\eta) \subseteq P(\eta + \xi) \subseteq C_\nu(\alpha)$, and thus, $\eta \in C_\nu(\alpha)$

We can additionally show that the ψ function only outputs additively principal ordinals:

Lemma 4

$\psi_\nu(\alpha) \in P$ for all ordinals ν, α.

Proof

Suppose $\psi_\nu(\alpha) \notin P$. Then $P(\psi_\nu(\alpha)) \subseteq \psi_\nu(\alpha)$, which is to say, for every element $p \in P(\psi_\nu(\alpha))$, $p < \psi_\nu(\alpha)$, which also means $p \in C_\nu(\alpha)$, thus $P(\psi_\nu(\alpha)) \subseteq C_\nu(\alpha)$. However, this implies $\psi_\nu(\alpha) \in C_\nu(\alpha)$, a contradiction.

Another interesting property of this function is that subscript of ψ represents the cardinality of it's result:

Proposition 18

$\Omega_\nu \leq \psi_{\nu(\alpha)} < \Omega_{\nu+1}$

Proof

Since $\Omega_\nu \subseteq C_\nu(\alpha)$, we get $\Omega_\nu \leq \psi_\nu(\alpha)$. Since the cardinality of $C_\nu(\alpha)$ is less than $\Omega_{\nu+1}$, we have $\psi_\nu(\alpha) < \Omega_{\nu+1}$.

So ψ_0 will only spit out countable ordinals $1 \leq \psi_0(\alpha) < \Omega$, ψ_1 will only be in the range $\Omega \leq \psi_1 < \Omega_2$, etc....

2 Bigger ordinals

Now let's look at $\psi_0(2)$. Since $C_0(1) \subseteq C_0(2)$, we will mainly focus on what *additional* elements are contained with $C_0(2)$. For starters, since $1 < 2$, we can access $\psi_0(1) = \omega, \psi_1(1), \ldots$. Since we can also use addition, ordinals like $\omega + 1$, $\omega + \omega$, or even $\omega \cdot 1000$ are within $C_0(2)$. However, as we cannot add ω to itself transfinitely many times, the least ordinal not contained within $C_0(2)$ is $\psi_0(2) = \omega^2$.

So far, as we have increased α, $\psi_0(\alpha)$ has increased. In fact, this is true for any ψ_ν:

Lemma 5

1. If $\alpha \leq \beta$, then $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\psi_\nu(\alpha) \leq \psi_\nu(\beta)$
2. If $\alpha < \beta$ and $\alpha \in C_\nu(\alpha)$, then $\psi_\nu(\alpha) < \psi_\nu(\beta)$

Proof

1. Trivial
2. From the previous statement, we can conclude that $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\psi_\nu(\alpha) \leq \psi_\nu(\beta)$. Since $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\alpha \in C_\nu(\alpha)$, $\alpha \in C_\nu(\beta)$, and since $\alpha < \beta$, $\psi_\nu(\alpha) \in C_\nu(\beta)$

Lemma 6

If $\gamma = \psi_{\nu_1}(\alpha_1) = \psi_{\nu_2}(\alpha_2)$, then $\nu_1 = \nu_2$ and $\alpha_1 = \alpha_2$, i.e., all outputs of ψ are uniquely represented.

Proof

Left as an exercise to the reader

Continuing from where we left off, we can repeat this for $C_0(3)$ to get $\psi_0(3) = \omega^3$, $\psi_0(4) = \omega^4$ and so on. It seems like $\psi_0(\alpha) = \omega^\alpha$, and we can prove that fact: We first prove a property about how $\psi_\nu(\alpha + 1)$ relates to $\psi_\nu(\alpha)$:

Lemma 7

1. $\psi_\nu(\alpha + 1) = \min\{\gamma \in P \mid \psi_\nu(\alpha) < \gamma\}$ if $\alpha \in C_\nu(\alpha)$
2. $\psi_\nu(\alpha + 1) = \psi_\nu(\alpha)$ if $\alpha \notin C_\nu(\alpha)$
3. If α is a limit ordinal, $\psi_\nu(\alpha) = \sup^{+}\{\psi_\nu(\xi) \mid \xi < \alpha \text{ and } \xi \in C_\nu(\xi)\}$

Proof

[WIP]

1. Since $\alpha \in C_\nu(\alpha)$, and $\alpha \in C_\nu(\alpha)$, $\psi_\nu(\alpha) < \psi_\nu(\alpha+1)$. Suppose for contradiction there exists an ordinal γ such that $\psi_\nu(\alpha) \leq \gamma < \psi_\nu(\alpha+1)$, and $\gamma \in P$.

Lemma 8

1. For all ordinals $\alpha < \varepsilon_0$, $\alpha \in C_0(\alpha)$ and $\psi_0(\alpha) = \omega^\alpha$
2. For all ordinals $\alpha < \varepsilon_{\Omega_\nu+1}$ where $\nu \neq 0$, $\alpha \in C_\nu(\alpha)$ and $\psi_\nu(\alpha) = \omega^{\Omega_\nu+\alpha}$

Proof

Suppose $\alpha \in C_0(\alpha)$ and $\psi_0(\alpha) = \omega^\alpha$. Then $\alpha+1 \in C_0(\alpha+1)$ and $\psi_0(\alpha+1)$

Using the lemma above, we have:

$$\begin{aligned}
\psi_0(1) &= \omega \\
\psi_0(2) &= \omega^2 \\
\psi_0(\omega) &= \omega^\omega \\
\psi_0(\omega+1) &= \omega^{\omega+1} \\
\psi_0(\omega^\omega) &= \omega^{\omega^\omega}
\end{aligned}$$

Then what about $\psi_0(\varepsilon_0)$?

$$\begin{aligned}
\psi_0(\varepsilon_0) &= \overset{+}{\sup}\{\psi_0(\xi) \mid \xi < \varepsilon_0 \text{ and } \xi \in C_0(\xi)\} \\
&= \left\{\omega^\xi \mid \xi < \varepsilon_0\right\} \\
&= \varepsilon_0
\end{aligned}$$

Therefore we have $\psi_0(\varepsilon_0) = \varepsilon_0$, and $\varepsilon_0 \notin C_0(\varepsilon_0)$. As such, we can't use ε_0 in $C_0(\varepsilon_0+1)$ (i.e., $\psi_0(\varepsilon_0) = \varepsilon_0 \notin C_0(\varepsilon_0+1)$), and $C_0(\varepsilon_0+1) = C_0(\varepsilon_0+1)$. So $\psi_0(\varepsilon_0+1) = \psi(\varepsilon_0) = \varepsilon_0$ In fact, none of the ordinals we have mentioned will cut it, $\psi_0(\Gamma_0) = \psi_0(\text{SVO}) = \psi_0(\text{LVO}) = \varepsilon_0$. This is where $\psi_1()$ comes in.

We (now) know that $\psi_1(0) = \Omega_1$. If you recall from the start, our $C_0(1)$ contains not only $\psi_0(0) = 1$, but also $\psi_1(0) = \Omega_1, \psi_2(0) = \Omega_2, \ldots$. From now on, we will use Ω as a shorthand for Ω_1. $\varepsilon_0 \notin C_0(\Omega)$, so $\psi_0(\Omega) = \varepsilon_0$. But since $\Omega \in C_0(\Omega) \subseteq C_0(\Omega+1)$, we **can** use $\psi_0(\Omega)$ in $C_0(\Omega+1)$, i.e., $\varepsilon_0 \in C_0(\Omega+1)$, so we have:

$$\begin{aligned}
\psi_0(\Omega) &= \varepsilon_0 \\
\psi_0(\Omega+1) &= \varepsilon_0 \cdot \omega = \omega^{\varepsilon_0+1} \\
\psi_0(\Omega+2) &= \varepsilon_0 \cdot \omega^2 = \omega^{\varepsilon_0+2} \\
\psi_0(\Omega+\omega) &= \varepsilon_0 \cdot \omega^\omega = \omega^{\varepsilon_0+\omega} \\
\psi_0(\Omega+\varepsilon_0) &= \varepsilon_0 \cdot \omega^{\varepsilon_0} = \varepsilon_0^2 = \omega^{\varepsilon_0+\varepsilon_0}
\end{aligned}$$

We can actually prove that $\psi_0(\Omega+\alpha) = \omega^{\varepsilon_0+\alpha}$:

Proof

We use transfinite induction:

1. Base case: $\psi_0(\Omega+0) = \omega^{\varepsilon_0+0} = \varepsilon_0$ and $0 \in C_0(\Omega+0)$

2. Successor ordinal case: If $\psi_0(\Omega+\alpha) = \omega^{\varepsilon_0+\alpha}$ and $\alpha \in C_0(\Omega+\alpha)$, then

$$\begin{aligned}\psi_0(\Omega+\alpha+1) &= \min\{\gamma \in P \mid \psi_0(\alpha) < \gamma\} \\ &= \min\left\{\omega^\xi \mid \omega^{\varepsilon_0+\alpha} < \omega^\xi\right\} \\ &= \omega^{\varepsilon_0+\alpha+1}\end{aligned}$$

   Since $\alpha, 1 \in C_0(\alpha) \subseteq C_0(\alpha+1)$, $\alpha+1 \in C_0(\alpha+1)$
3. Limit ordinal case: If for a limit ordinal λ, for all ordinals $\alpha < \lambda$, $\psi_0(\Omega+\alpha) = \omega^{\varepsilon_0+\alpha}$ and $\alpha \in C_0(\Omega+\alpha)$,

$$\begin{aligned}\psi_0(\Omega+\lambda) &= \overset{+}{\sup}\{\psi_0(\Omega+\alpha) \mid \alpha < \lambda \text{ and } \alpha \in C_0(\alpha)\} \\ &= \overset{+}{\sup}\{\omega^{\varepsilon_0+\alpha} \mid \alpha < \lambda\} \\ &= \omega^{\varepsilon_0+\lambda}\end{aligned}$$

   However, we need to show that $\Omega+\lambda \in C_0(\Omega+\lambda)$ in order to *continue* the induction to higher ordinals. If $\Omega+\lambda \in C_0(\Omega+\lambda)$, then $\lambda \in C_0(\Omega+\lambda)$ and therefore we must have $\lambda < \psi_0(\Omega+\lambda)$.

   Since $\psi_0(\Omega+\lambda) = \omega^{\varepsilon_0+\lambda}$, $\lambda < \omega^{\varepsilon_0+\lambda}$. The first ordinal that violates this property is the *second* fixed point of $\omega^\bullet = \bullet$, i.e., ε_1. Though note that for ε_1, $\psi_0(\Omega+\varepsilon_1) = \omega^{\varepsilon_0+\varepsilon_1} = \varepsilon_1$ still holds, just that $\varepsilon_1 \notin C_0(\Omega+\varepsilon_1)$

So therefore, for all $\alpha \leq \varepsilon_1$, $\psi_0(\Omega+\alpha) = \omega^{\varepsilon_0+\alpha}$

We see that our function gets "stuck" once again on $\psi_0(\Omega+\varepsilon_1)$. It only becomes "unstuck" once again when we have an ordinal $\alpha > \Omega+\varepsilon_1$ such that $\alpha \in C_0(\alpha)$. The next such ordinal is $\Omega+\Omega$, or $\Omega \cdot 2$, and we have $\psi_0(\Omega \cdot 2) = \varepsilon_1$.

We can similarly have

$$\begin{aligned}\psi_0(\Omega \cdot 3) &= \varepsilon_2 \\ \psi_0(\Omega \cdot \omega) &= \varepsilon_\omega \\ \psi_0(\Omega \cdot \varepsilon_0) &= \varepsilon_{\varepsilon_0} \\ \psi_0(\Omega \cdot \Omega) &= \zeta_0\end{aligned}$$

, replacing each "stuck" term with Ω, and in general, we can relate Buchholz's function to the Veblen function as such:

Proposition 19

$$\psi_0(\Omega^\alpha \cdot (1+\beta)) = \varphi(\alpha, \beta)$$

Proof

[WIP] probably shift to some appendix cause its quite long I think.

We can generalize this to the extended veblen function as such:

Proposition 20

$$\psi_0\left(\Omega^{(\Omega^n\alpha_n+\ldots+\Omega\alpha_1+\alpha_0)}(1+\beta)\right) = \varphi(\alpha_n, \ldots, \alpha_1, \alpha_0, \beta)$$

You can verify that these expressions containing Ω satisfies $\alpha \in C_0(\alpha)$ by expressing them in terms of ψ_1:

$$\begin{aligned}
\psi_1(0) &= \Omega \\
\psi_1(\alpha) &= \omega^{\Omega+\alpha} = \Omega \cdot \omega^\alpha \\
\psi_1(\psi_1(0)) &= \psi_1(\Omega) = \omega^{\Omega+\Omega} = \Omega^2 \\
\psi_1(\psi_1(\alpha)) &= \psi_1(\Omega \cdot \omega^\alpha) \\
\psi_1(\Omega \cdot \alpha) &= \omega^{\Omega+\Omega\cdot\alpha} = \omega^{\Omega\cdot\alpha} = \Omega^\alpha \\
\psi_1(\Omega^2) &= \Omega^\Omega
\end{aligned}$$

So for example:

$$\begin{aligned}
\zeta_0 &= \psi_0(\Omega^2) \\
&= \psi_0(\psi_1(\Omega)) \\
&= \psi_0(\psi_1(\psi_1(0)))
\end{aligned}$$

The limit of the 2-argument veblen function is Γ_0, which can be written as:

$$\begin{aligned}
\Gamma_0 &= \psi_0(\Omega^\Omega) \\
&= \psi_0(\psi_1(\Omega^2)) \\
&= \psi_0(\psi_1(\psi_1(\Omega))) \\
&= \psi_0(\psi_1(\psi_1(\psi_1(0))))
\end{aligned}$$

Continuing, we have

$$\text{SVO} = \psi_0(\Omega^{\Omega^\omega}), \text{LVO} = \psi_0\left(\Omega^{\Omega^\Omega}\right)$$

Now that we have "caught up" with the veblen function, we can go even further, with $\psi_0\left(\Omega^{\Omega^{\Omega^\Omega}}\right)$. Eventually we reach $\psi_0(\Omega^{\Omega^{\cdot^{\cdot^{\cdot}}}}) = \psi_0(\varepsilon_{\Omega+1})$, the **Bachmann-Howard Ordinal (BHO)**. (The expression $\varepsilon_{\Omega+1}$ really just means the supremum of $\left\{\Omega, \Omega^\Omega, \Omega^{\Omega^\Omega}, \ldots\right\}$).

So far we've been using ψ_1 to construct all our terms containing Ω. However, just like how we previously got "stuck" at $\psi_0(\varepsilon_0) = \varepsilon_0$, we run into the issue where $\psi_1(\varepsilon_{\Omega+1}) = \varepsilon_{\Omega+1}$:

$$\begin{aligned}
\psi_1(\varepsilon_{\Omega+1}) &= \overset{+}{\sup}\left(\psi_1(\Omega), \psi_1(\Omega^\Omega), \psi_1\left(\Omega^{\Omega^\Omega}\right), \ldots\right) \\
&= \overset{+}{\sup}\left\{\omega^{\Omega+\Omega}, \omega^{\Omega+\Omega^\Omega}, \omega^{\Omega+\Omega^{\Omega^\Omega}}, \ldots\right\} \\
&= \overset{+}{\sup}\left\{\Omega^2, \Omega^\Omega, \Omega^{\Omega^\Omega}, \ldots\right\} \\
&= \varepsilon_{\Omega+1}
\end{aligned}$$

Since $\varepsilon_{\Omega+1} \notin C_1(\varepsilon_{\Omega+1})$ and $C_0(\varepsilon_{\Omega+1}) \subset C_1(\varepsilon_{\Omega+1})$, $\varepsilon_{\Omega+1} \notin C_0(\varepsilon_{\Omega+1})$. To go further than this, we will need to invoke higher Ωs, like Ω_2, Ω_3, and so on. The next ordinal larger than $\varepsilon_{\Omega+1}$ that we can access is $\psi_2(0) = \Omega_2$, so we have $\psi_1(\Omega_2) = \varepsilon_{\Omega+1}$, and can re-express the Bachmann-Howard Ordinal as $\psi_0(\Omega_2)$ instead.

But wait, if we have $\text{BHO} = \psi_0(\Omega_2)$ and $\text{BHO} = \psi_0(\varepsilon_{\Omega+1}) = \psi_0(\psi_1(\Omega_2))$, which is the standard way to write it? As we have seen earlier, having the property $\alpha \in C_\nu(\alpha)$ saves us a lot of headaches by avoiding places where the function "gets stuck". So $\psi_0(\Omega_2)$ is "nicer" as $\Omega_2 \in C_0(\Omega_2)$, whereas $\psi_0(\varepsilon_{\Omega+1}) = \psi_0(\psi_1(\Omega_2))$ is "less nice" as $\varepsilon_{\Omega+1} \notin C_0(\varepsilon_{\Omega+1})$. We'll elaborate more on this when we go over the Normal Form for Buchholz's ψ. From now on, in this section, we'll color terms where $\alpha \notin C_\nu(\alpha)$ and the values that they get "stuck" at in purple.

We then have:

$$\begin{aligned}
\psi_0(\Omega_2 + 1) &= \text{BHO} \cdot \omega \\
\psi_0(\Omega_2 + \omega) &= \text{BHO} \cdot \omega^\omega \\
\psi_0(\Omega_2 + \text{BHO}) &= \text{BHO} \cdot \omega^{\text{BHO}} = \text{BHO}^2 \\
\psi_0(\Omega_2 + \text{BHO}^2) &= \text{BHO} \cdot \omega^{\text{BHO}^2} = \text{BHO}^{\text{BHO}} \\
\psi_0\left(\Omega_2 + \text{BHO}^{\text{BHO}}\right) &= \text{BHO} \cdot \omega^{\text{BHO}^{\text{BHO}}} = \text{BHO}^{\text{BHO}^{\text{BHO}}} \\
\psi_0(\Omega_2 + \varepsilon_{\text{BHO}+1}) &= \varepsilon_{\text{BHO}+1}
\end{aligned}$$

Just like how $\psi_0(\Omega + \varepsilon_1) = \varepsilon_1$, we have $\psi_0(\Omega_2 + \varepsilon_{\text{BHO}+1}) = \varepsilon_{\text{BHO}+1}$, so we replace it with an Ω and get $\psi_0(\Omega_2 + \Omega) = \varepsilon_{\text{BHO}+1}$. With this we can find the next "fixed point":

$$\begin{aligned}
\psi_0(\Omega_2 + \Omega) &= \varepsilon_{\text{BHO}+1} \\
\psi_0(\Omega_2 + \Omega + \varepsilon_{\text{BHO}+1}) &= \varepsilon_{\text{BHO}+1}^2 \\
\psi_0\left(\Omega_2 + \Omega + \varepsilon_{\text{BHO}+1}^2\right) &= \varepsilon_{\text{BHO}+1}^{\varepsilon_{\text{BHO}+1}} \\
\psi_0\left(\Omega_2 + \Omega + \varepsilon_{\text{BHO}+1}^{\varepsilon_{\text{BHO}+1}}\right) &= \varepsilon_{\text{BHO}+1}^{\varepsilon_{\text{BHO}+1}^{\varepsilon_{\text{BHO}+1}}} \\
\psi_0(\Omega_2 + \Omega + \varepsilon_{\text{BHO}+2}) &= \varepsilon_{\text{BHO}+2} \\
\psi_0(\Omega_2 + \Omega + \Omega) &= \varepsilon_{\text{BHO}+2}
\end{aligned}$$

This is starting to look like the veblen correspondence all over again, where

$$\psi_0(\Omega_2 + \Omega^\alpha \cdot \beta) = \varphi(\alpha, \text{BHO} + \beta)$$

Now let's look at ψ_1:

$$\begin{aligned}
\psi_1(\Omega_2) &= \varepsilon_{\Omega+1} \\
\psi_1(\Omega_2 + 1) &= \varepsilon_{\Omega+1} \cdot \omega \\
\psi_1(\Omega_2 + \varepsilon_{\Omega+1}) &= \varepsilon_{\Omega+1} \cdot \omega^{\varepsilon_{\Omega+1}} = \varepsilon_{\Omega+1}^2 \\
\psi_1\left(\Omega_2 + \varepsilon_{\Omega+1}^2\right) &= \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}} \\
\psi_1\left(\Omega_2 + \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}}\right) &= \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}}} \\
\psi_1(\Omega_2 + \varepsilon_{\Omega+2}) &= \varepsilon_{\Omega+2}
\end{aligned}$$

So we have to yet again find the next ordinal $\alpha > \Omega_2 + \varepsilon_{\Omega+2}$ such that $\alpha \in C_1(\alpha)$. The first ordinal to satisfy that criteria is $\alpha = \Omega_2 \cdot 2$, so we get $\psi_1(\Omega_2 \cdot 2) = \varepsilon_{\Omega+2}$. We again have a similar correspondence going on:

$$\psi_1(\Omega_2^\alpha \cdot \beta) = \varphi(\alpha, \Omega + \beta)$$

But we don't need the veblen function anymore. The limit of what we can do with Ω_2 is:

$$\psi_1\left(\Omega_2^{\Omega_2^{\cdot^{\cdot^{\cdot}}}}\right) = \psi_1\left(\varepsilon_{\Omega_2+1}\right)$$

This is analogous the BHO earlier, where instead of $\psi_1(\varepsilon_{\Omega+1}) = \varepsilon_{\Omega+1}$, we have $\psi_2\left(\varepsilon_{\Omega_2+1}\right) = \varepsilon_{\Omega_2+1}$ (Convince yourself that this is true). So we get $\psi_1(\Omega_3) = \varepsilon_{\Omega_2+1}$, and $\psi_0\left(\varepsilon_{\Omega_2+1}\right) = \psi_0(\Omega_3)$.

We can keep going $\psi_0(\Omega_4), \psi_0(\Omega_5), \ldots$ up to $\psi_0(\Omega_\omega)$, which is known as **Buchholz's Ordinal (BO)**. This is the limit of all $\psi_0(\Omega_n)$ for finite n, and while we haven't gotten to the ordinal notation associated with Buchholz's ψ yet, there are many ordinal notations that correspond to ordinals below $\psi_0(\Omega_\omega)$.

Of course, the true limit of Buchhol'z OCF is $\psi_0\left(\Omega_\omega^{\Omega_\omega^{\Omega_\omega^{\cdot^{\cdot^{\cdot}}}}}\right) = \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$, the **Takeuti–Feferman–Buchholz ordinal (TFBO)**, but it is not as commonly used.

3 Normal Form and Fundamental Sequences

Theorem 8

Every ordinal $\alpha < \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$ can be uniquely expressed in the form:

$$\alpha = \psi_{k_1}(\alpha_1) + \ldots + \psi_{k_n}(\alpha_n)$$

Where:
- $\alpha_1, \ldots, \alpha_n \in C_0\left(\varepsilon_{\Omega_\omega+1}\right)$
- $k_1, \ldots, k_n \leq \omega$
- $\psi_{k_1}(\alpha_1) \geq \ldots \geq \psi_{k_n}(\alpha_n)$
- $\alpha_1 \in C_{k_1}(\alpha_1), \ldots, \alpha_n \in C_{k_n}(\alpha_n)$

Like in Cantor Normal Form, α_i does not necessarily have to be in normal form.

Example 38

Which of the following are in normal form:
1. $\psi_0(0)$
2. $\psi_0(\Omega)$
3. $\psi_0(\psi_0(\Omega))$
4. $\psi_0(\psi_1(\psi_2(\Omega_3)))$

Answers:
1. $0 \in C_0(0)$, normal form
2. $\Omega = \psi_1(0) \in C_0(\Omega)$, normal form
3. $\psi_0(\Omega) = \varepsilon_0 \notin C_0(\varepsilon_0)$, not normal form

Note that just because an ordinal α is expressed in normal form, that does not mean $\psi_\nu(\alpha)$ is in normal form. ($\psi_0(\Omega)$ is normal form but $\psi_0(\psi_0(\Omega))$ isn't)

1. Firstly we have $\psi_2(\Omega_3) = \varepsilon_{\Omega_2+1}$. Then we have $\psi_1(\psi_2(\Omega_3)) = \psi_1\left(\varepsilon_{\Omega_2+1}\right)$, which in normal form is $\psi_1(\Omega_3)$. We then construct the following inequality to solve for $\psi_0(\psi_1(\Omega_3))$:

$$\psi_1(\Omega_2) \leq \psi_1(\Omega_3) < \Omega_2$$
$$\psi_0(\psi_1(\Omega_2)) \leq \psi_0(\psi_1(\Omega_3)) \leq \psi_0(\Omega_2)$$
$$\text{BHO} \leq \psi_0(\psi_1(\Omega_3)) \leq \text{BHO}$$

Therefore $\psi_0(\psi_1(\Omega_3)) = \text{BHO}$. However the BHO has only one unique normal form expression: $\psi_0(\Omega_2)$. So $\psi_0(\psi_1(\psi_2(\Omega_3))) = \psi_0(\psi_1(\Omega_3))$ are both not normal form.

Before we go into the fundamental sequences of ψ, we first introduce a concept known as **cofinality**:

Definition 55

The **cofinality** of an ordinal α, denoted cof α, is the **least cardinal number** κ such that there exists a subset $S \subseteq \alpha$ of cardinality κ where α is smallest ordinal strictly greater than every element in S.

Example 39

Find the cofinality of
1. $\omega 2$
2. $\omega 2 + 3$

Answers:
1. Let $S = \{\omega + n \mid n \in \mathbb{N}\} \subset \omega 2$. S has a cardinality of ω.

   However, we still need to rule out the possibility that there might be a subset S of finite cardinality (i.e. cardinality $< \omega$) that still satisfies the criteria that $\omega 2$ is the smallest ordinal greater than every element in S

   In a subset S which has a nonzero finite cardinality, there is a largest element β, and its successor $\beta + 1$ is the smallest ordinal greater than all elements of the subset S while being less than $\omega 2$. Therefore S can't have nonzero finite cardinality.

   If S has a cardinality of 0 (i.e. $S = \emptyset = \{\}$), then 0 is the smallest ordinal greater than all elements in S. Therefore S can't have a cardinality of 0.

   Therefore the smallest cardinality of S where $\omega 2$ is the least ordinal greater than all elements in S is ω.
2. $S = \{\omega 2 + 2\} \subset \omega 2 + 3$, and S has a cardinality of 1. S can't have a cardinality of 0 since the smallest ordinal larger than nothing is 0.

Now onto fundamental sequences. A fundamental sequence for an ordinal number α with cofinality $\text{cof}(\alpha) = \beta$ is a strictly increasing sequence $(\alpha[0], \alpha[1], \ldots)$ with length β and limit α (i.e. $\alpha = \overset{+}{\sup}\{\alpha[\eta] \mid \eta < \beta\}$)

[WIP: Other systems of fundamental sequences]

Definition 56 (**Denis Maksudov's system of fundamental sequences**)

For ordinals $\alpha \leq \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$ (expressed in normal form), the ηth term of the fundamental sequence of α, $\alpha[\eta]$ is defined as such:
1. If $\alpha = \psi_{k_1}(\alpha_1) + \ldots + \psi_{k_n}(\alpha_n)$, then $\text{cof}(\alpha) = \text{cof}\left(\psi_{k_n}(\alpha_n)\right)$, and $\alpha[\eta] = \psi_{k_1}(\alpha_1) + \ldots + \left(\psi_{k_n}(\alpha_n)[\eta]\right)$
2. If $\alpha = \psi_0(0) = 1$ then $\text{cof}(\alpha) = 1$. We need a sequence of length 1, so we set $\alpha[0] = 0$
3. If $\alpha = \psi_{\nu+1}(0) = \Omega_{\nu+1}$, then $\text{cof}(\alpha) = \Omega_{\nu+1}$. We need a sequence of length $\Omega_{\nu+1}$, so we set $\alpha[\eta] = \Omega_{\nu+1}[\eta] = \eta$

4. If $\alpha = \psi_\omega(0)$, then $\mathrm{cof}(\alpha) = \mathrm{cof}(\omega) = \omega$ and $\alpha[\eta] = \psi_{\nu[\eta]}(0) = \Omega_{\nu[\eta]}$
5. If $\alpha = \psi_\nu(\beta + 1)$ then $\mathrm{cof}(\alpha) = \omega$ and $\alpha[\eta] = \psi_\nu(\beta) \cdot \eta$
6. If $\alpha = \psi_\nu(\beta)$ where β is a limit ordinal whose cofinality $\mathrm{cof}(\beta) \in \{\omega\} \cup \{\Omega_{\mu+1} \mid \mu < \nu\}$, then we have $\mathrm{cof}(\alpha) = \mathrm{cof}(\beta)$, so we need a sequence of length $\mathrm{cof}(\beta)$, so $\alpha[\eta] = \psi_\nu(\beta[\eta])$
7. If $\alpha = \psi_\nu(\beta)$ where β is a limit ordinal whose cofinality $\mathrm{cof}(\beta) = \Omega_{\mu+1}$ for $\mu \geq \nu$, then $\mathrm{cof}(\alpha) = \omega$ and $\alpha[\eta] = \psi_\nu(\beta[\gamma[n]])$ where $\gamma[0] = \Omega_\mu$, and $\gamma[\eta + 1] = \psi_\nu(\beta[\gamma[\eta])$

There's a lot to go through here, so let's go through some examples:

Example 40

1.

4 Extended Buchholz's function

Why do we have to stop at Ω_ω? What about $\Omega_{\omega+1}$? We can extend Buchholz's function as such to allow for even larger ordinals:

Definition 57 (**Extended Buchholz's function**)

Using the same definition of Ω_ν, we define the **Extended Buchholz's function** ψ and the set $C_\nu(\alpha)$ as such:
1. $\Omega_\nu \subseteq C_\nu(\alpha)$
2. For any two ordinals $\xi, \eta \in C_\nu(\alpha)$, their sum $\xi + \eta \in C_\nu(\alpha)$
3. For any ordinal $\xi \in C_\nu(\alpha)$, as long as $\xi < \alpha$, then $\psi_\mu(\xi) \in C_\nu(\alpha)$ for all $\mu \in C_\nu(\alpha)$
4. $\psi_\nu(\alpha) = \min\{\gamma \in \mathrm{Ord} \mid \gamma \notin C_\nu(\alpha)\}$, i.e., the smallest ordinal not inside $C_\nu(\alpha)$

It's the same as the original, just that instead of being restricted to ψ_μ with $\mu \leq \omega$, we now have $\mu \in C_\nu(\alpha)$ as highlighted in red.

With this, we can go beyond, letting $\mathrm{TFBO} = \psi_0(\Omega_{\omega+1})$. We can make $\psi_0\left(\Omega_{\varepsilon_0}\right)), \psi_0\left(\Omega_{\Gamma_0}\right)$, or even $\psi_0(\Omega_\Omega)$.

To

09 | Ordinal Notations up to $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$

We introduce a set of ordinal notations OT along with a recursive way to order them $\prec$, such that $(OT, \prec)$ is order isomorphic to $\left(C_0\left(\varepsilon_{\Omega_\omega+1}\right), <\right)$

Definition 58

Let T be the set of terms and PT be the set of all prinicpal terms (the terms in PT are associated to additively principal ordinals).

- $\mathbf{0} \in T$
- Given a principal term $s \in PT$ and a term $t \in T$, $s + t \in T$
- Given a term $t \in T$, $\mu \leq \omega$, $\psi_\mu(t) \in T$ and PT

Let $\prec$ be a binary relation on T.

For terms $s, t \in T$, $s \prec t$ is defined as:
- If $s = \mathbf{0}$ then $s \prec t$ if $t \neq \mathbf{0}$
- If $t = \mathbf{0}$ then $s \prec t$ is false
- If $s = a + b$ where $a \in PT$ and $b \in T \setminus \{\mathbf{0}\}$, then
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{\mathbf{0}\}$
    - If $a \neq c$ then $s \prec t$ if and only if $a \prec c$
    - If $a = c$ then $s \prec t$ if and only if $b \prec d$
  - If $t \in PT$ then $s \prec t$ if and only if $a \prec t$
- If $s = \psi_\mu(a)$ for some $a \in T, \mu \leq \omega$ then:
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{\mathbf{0}\}$, then $s \prec t$ if and only if $s \preceq c$
  - If $t = \psi_\nu(b)$ for some $b \in T, \nu \leq \omega$ then:
    - If $\mu \neq \nu$ then $s \prec t$ if and only if $\mu \prec \nu$
    - If $\mu = \nu$ then $s \prec t$ if and only if $a \prec b$

For terms s, t, we have a family of the binary relations $s \in C_\mu(t)$ for $\mu \leq \omega$ if and only if:

- $s \prec \psi_\mu(\mathbf{0})$
- $s = a + b$ for some $a \in PT$ and $b \in T \setminus \{\mathbf{0}\}$ and $a \in C_\mu(t)$ and $b \in C_\mu(t)$
- $s = \psi_\nu(a)$ for some $a \in T, \nu \leq \omega$ and $a \in C_\mu(t)$ and $b \prec t$

Let OT be a subset of T. OT corresponds to the set of all ordinal notations of normal form. OT is defined as:

- $\mathbf{0} \in OT$
- If $a_0, \ldots, a_k \in OT$ and $a_0, \ldots, a_k \in PT$, if $a_k \preceq \ldots \preceq a_0$, their sum $a_0 + \ldots + a_k \in OT$
- For a term $a \in T$, $\psi_\mu(a) \in OT$ if and only if $a \in OT$ and $a \in C_\mu(a)$

We can now define a map to associate each ordinal notation to an ordinal below $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$:

Definition 59

The map $o : OT \mapsto C_0\left(\varepsilon_{\Omega_\omega+1}\right)$ is defined recursively as such:

1. $o(\mathbf{0}) := 0$

2. $o(a_0 + \ldots + a_k) = o(a_0) + \ldots + o(a_k)$
3. $o(\psi_\nu(a)) = \psi_\nu(o(a))$

This should be fairly self-explanatory with the notation that has been chosen essentially being red versions of the usual ordinals.

In Buchholz's original paper, where o was originally defined, he formally proved that $(OT, \prec)$ is order isomorphic to $\left(C_0\left(\varepsilon_{\Omega_\omega+1}\right), <\right)$, that they are well-ordered (no infinite descending sequence), and that these two sets have order type $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$. Note that he used $D_\nu a$ instead of our red $\psi_\nu(a)$, and $G_\nu(b)$ is somewhat analogous to $C_\nu(b)$.

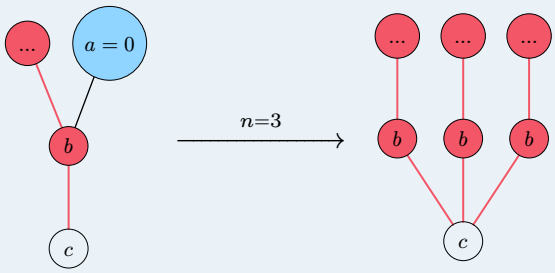
1 Buchholz's Hydra

Definition 60

This is somewhat similar to the Kirby-Paris hydra game we have earlier, but each node is labelled with an ordinal $\leq \omega$, the child nodes of the root must be labelled 0.

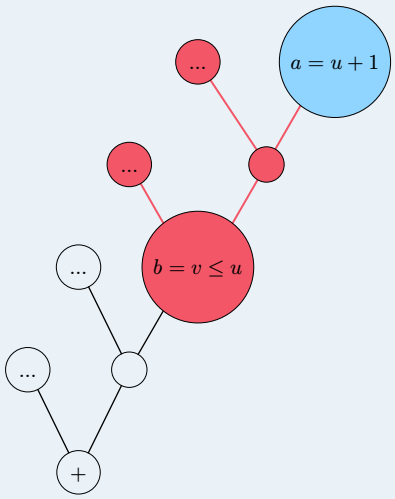
At each step, we choose a leaf node a to chop off. The hydra chooses some $n \in \mathbb{N}$.

1. If a has label 0, we proceed as in Kirby-Paris' game. Call the node's parent b, and its grandparent c (if it exists). First we delete a. If c exists (i.e. b is not the root), we make n copies of b and all its children and attach them to c:

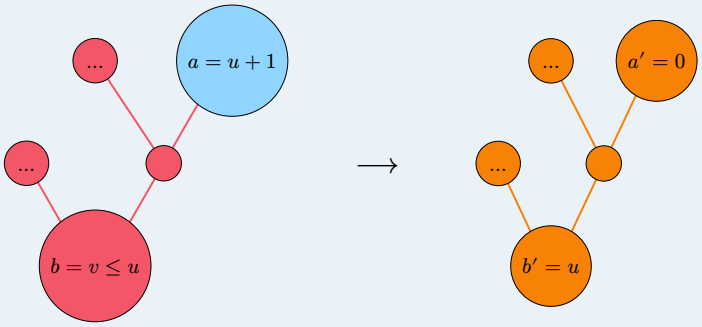


2. If a has label $u + 1$, we go down the tree looking for a node b whose label is $v \leq u$. Let the subtree rooted in b be called S. We make a copy of S called S'. We re-label the b inside S' to u and re-label the a inside S' to be 0. We then replace a in the original tree with S'

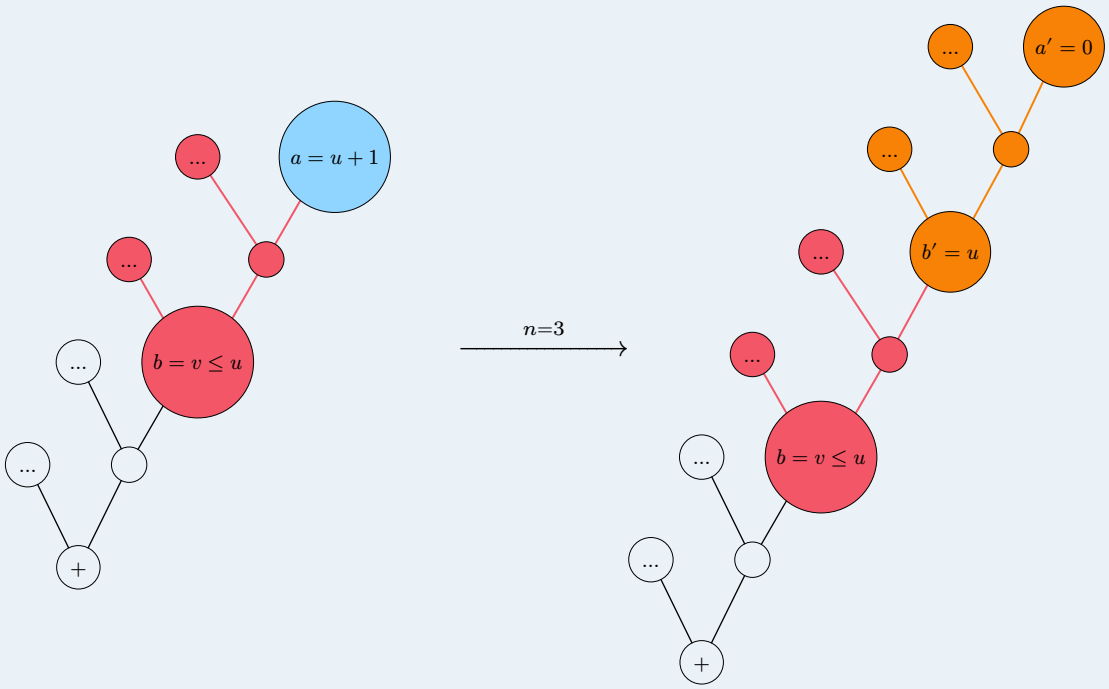
   Here's an illustration. If this is the hydra:



then the nodes highlighted in red (and a highlighted in blue) make up S. We then make a copy called S' and make the changes accordingly:



Then replacing the original a with S' gives:



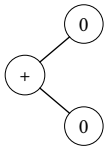
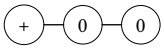
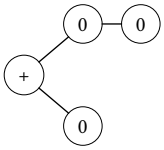
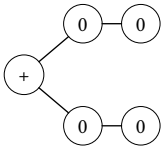
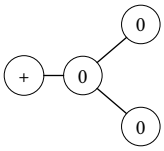
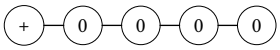
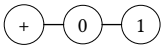
3. If a has label ω, replace the label ω with $n + 1$

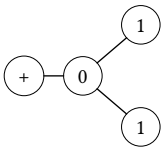
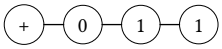
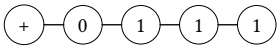
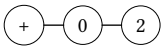
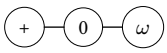
Similarly to the Kirby-Paris Hydra, we can associate each hydra with an ordinal notation by the following rules:

1. For each leaf node with a label u, we replace it with the ordinal notation $\psi_u(0)$
2. For a node with label u with children of ordinal notations $a_0, \ldots, a_k$, we replace it with the ordinal notation $\psi_u(a_0 + \ldots + a_k)$
3. At the root node, the resulting ordinal expressions of each of the root's children $a_0, \ldots, a_k$ are summed up: $a_0 + \ldots + a_k$

Here are some examples of Buchholz's hydra and the associated ordinal notations and ordinals. The hydras have been rotated to save space.

Buchholz Hydra	Associated Ordinal Notation	Associated Ordinal
+	0	0

Buchholz Hydra	Associated Ordinal Notation	Associated Ordinal
	$\psi_0(\mathbf{0})$	1
	$\psi_0(\mathbf{0}) + \psi_0(\mathbf{0})$	2
	$\psi_0(\psi_0(\mathbf{0}))$	ω
	$\psi_0(\psi_0(\mathbf{0})) + \psi_0(\mathbf{0})$	$\omega + 1$
	$\psi_0(\psi_0(\mathbf{0})) + \psi_0(\psi_0(\mathbf{0}))$	$\omega 2$
	$\psi_0(\psi_0(\mathbf{0}) + \psi_0(\mathbf{0}))$	ω^2
	$\psi_0(\psi_0(\psi_0(\mathbf{0})))$	ω^ω
	$\psi_0(\psi_0(\psi_0(\psi_0(\mathbf{0}))))$	ω^{ω^ω}
	$\psi_0(\psi_1(\mathbf{0}))$	$\psi_0(\Omega) = \varepsilon_0$

Buchholz Hydra	Associated Ordinal Notation	Associated Ordinal
	$\psi_0(\psi_1(\mathbf{0}) + \psi_1(\mathbf{0}))$	$\psi_0(\Omega 2) = \varepsilon_1$
	$\psi_0(\psi_1(\psi_1(\mathbf{0})))$	$\psi_0(\Omega^2) = \zeta_0$
	$\psi_0(\psi_1(\psi_1(\psi_1(\mathbf{0}))))$	$\psi_0(\Omega^\Omega) = \Gamma_0$
	$\psi_0(\psi_1(\psi_1(\psi_1(\psi_1(\mathbf{0})))))$	$\psi_0\left(\Omega^{\Omega^\Omega}\right) = \mathrm{LVO}$
	$\psi_0(\psi_2(\mathbf{0}))$	$\psi_0(\Omega_2) = \mathrm{BHO}$
	$\psi_0(\psi_\omega(\mathbf{0}))$	$\psi_0(\Omega_\omega) = \mathrm{BO}$

2 The Hyper Primitive Sequence System (HPrSS)

3 Limited Patcail's Hydra

4 The Pair Sequence System (PSS)

The Pair Sequence System (PSS) can also be considered an extension to the primitive sequence system, where instead of it being a sequence of natural numbers, the Pair Sequence System can be thought og as a sequence of *pairs* of natural numbers. There are two common ways to write PSS, either inline, like $(0,0)(1,1)(2,1)(2,0)$ or as a matrix like so:

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

divergence at $\psi_0(\Omega_\omega \cdot (\Omega + 1))$

5 Ordinal notation associated with Extended Buchholz's function

Definition 61

Let T be the set of terms and PT be the set of all prinicpal terms (terms whose associated ordinal are additively principal ordinals).

- $0 \in T$
- Given a principal term $s \in PT$ and a term $t \in T$, $s + t \in T$
- Given terms $s, t \in T$, $\psi_s(t) \in T$ and PT

Let $\prec$ be a binary relation on T.

For terms $s, t \in T$, $s \prec t$ is defined as:

- If $s = 0$ then $s \prec t$ if $t \neq 0$
- If $t = 0$ then $s \prec t$ is false
- If $s = a + b$ where $a \in PT$ and $b \in T \setminus \{0\}$, then
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{0\}$
    - If $a \neq c$ then $s \prec t$ if and only if $a \prec c$
    - If $a = c$ then $s \prec t$ if and only if $b \prec d$
  - If $t \in PT$ then $s \prec t$ if and only if $a \prec t$
- If $s = \psi_a(b)$ for some $a, b \in T$ then:
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{0\}$, then $s \prec t$ if and only if $s \preceq c$
  - If $t = \psi_c(d)$ for some $c, d \in T$ then:
    - If $a \neq c$ then $s \prec t$ if and only if $a \prec c$
    - If $a = c$ then $s \prec t$ if and only if $b \prec d$

For terms s, t, u, we define the ternary relation $s \in C_t(u)$ if and only if:

- $s \prec \psi_t(0)$
- $s = a + b$ for some $a \in PT$ and $b \in T \setminus \{0\}$ and $a \in C_t(u)$ and $b \in C_t(u)$
- $s = \psi_a(b)$ for some $a, b \in T$ and $a \in C_t(u)$ and $b \in C_t(u)$ and $b \prec u$

Let OT be a subset of T. OT corresponds to the set of all ordinal notations of normal form. For a term s, $s \in OT$ is defined as:

- If $s = 0$ then $s \in OT$
- If $s = a + b$ for some $a, b \in PT$, then $s \in OT$ if and only if $a, b \in OT$ and $b \preceq a$
- If $s = a + b + c$ for some $a, b \in PT$ and $c \in T \setminus \{0\}$ then $s \in OT$ if and only if $a, (b + c) \in OT$ and $b \preceq a$
- If $s = \psi_a(b)$ for some $a, b \in T$ then $s \in OT$ if and only if $a, b \in OT$ and $b \in C_a(b)$

010 Bashicu matrix system (BMS)